

# The **posim** Command Language

## Complete Grammar & Notebook Guide

physical\_object\_simulator

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# 1 What is this document about?

**posim** is the front end of the `physical_object_simulator`. Instead of writing Rust code to create objects and run physics, you type short *commands* — one per line — into a *notebook*. Each command line is:

1. chopped into *tokens* by a *lexer* (like the classic Unix tool `lex/flex`);
2. checked against a *grammar* and compiled into a small program by a *parser* (like `yacc/bison`);
3. executed by a *stack machine* — a tiny computer whose only memory is a stack of values — which reads and writes the simulator exclusively through its public `get/set` functions.

You never see steps 1–3; you type a line, press Enter (or shift-enter in JupyterLab), and see the result. This document specifies exactly what you may type.

There are three ways to talk to the same language:

| mode                     | start with                                                       | who it is for                                                                             |
|--------------------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| notebook REPL            | <code>cargo run</code> (or <code>posim</code> )                  | humans at a terminal                                                                      |
| script                   | <code>posim -script file</code>                                  | batch runs, reproducibility                                                               |
| dynamic notebook machine | <code>posim -notebook file</code><br><code>posim -machine</code> | load a notebook, open its scene window, continue live programs, and the JupyterLab kernel |

## 2 Lexical structure (what the lexer sees)

A command line is a sequence of *tokens* separated by optional whitespace. Everything from a `#` to the end of the line is a *comment* and is ignored.

### 2.1 Token kinds

| token       | examples                                                                | rules                                                                                                                                                                            |
|-------------|-------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| keyword     | <code>NEW</code> , <code>set</code> , <code>Run</code>                  | case-insensitive; list in §2.2                                                                                                                                                   |
| identifier  | <code>obj0</code> , <code>position</code>                               | a letter or <code>_</code> , then letters, digits, <code>_</code>                                                                                                                |
| string      | <code>"ball"</code> , <code>"dumbell10"</code>                          | double-quoted, with the escapes <code>\"</code> <code>\\</code> <code>\n</code> ; strings name objects ( <code>NEW ... AS</code> , §5.1) and are passed to user functions (§5.9) |
| number      | <code>2</code> , <code>0.5</code> , <code>.5</code> , <code>1e-3</code> | 64-bit float; scientific notation allowed; a leading <code>-</code> is <i>not</i> part of the number — it is the negation operator                                               |
| punctuation | <code>[ ] { } ( ) , . =</code>                                          | brackets build vectors, braces enclose initializers, <code>.</code> builds dotted paths                                                                                          |
| operators   | <code>+</code> <code>-</code> <code>*</code> <code>/</code>             | ordinary arithmetic                                                                                                                                                              |

If the lexer meets a character it does not know (say `@`), it stops with an error naming the *column* (character position, starting at 1):

```
Err[1]: lexical error at column 10: unexpected character '@'
```

## 2.2 Keywords

NEW SET GET DEL DELETE LIST STEP RUN STEPS METHOD ADAMS BDF SPRK ENERGY COM MOMENTUM  
ANGMOM LAPLACE HELP POINT SPHERE CUBOID TORUS DISK CYLINDER BOX RESET

and, for the graphical scene window (documented in `scene_info.pdf`; Examples 11–12 show live sessions):

SCENE CREATE CLOSE DESTROY TRANSLATE ROTATE ZOOM IN OUT HIDE SHOW REFRESH REDRAW START  
STOP PAUSE REVERSE SET\_TIME\_STEP SETTIMESTEP STATUS EVENTS ALL

and, for rigid-body collisions (§5.7):

COLLIDE CONTACTS ON OFF

and, for the quantum families (§5.10, §5.11, §5.12):

QM QM2 QM3

and, for named objects, session variables and user functions (§5.1, §5.9):

AS LET FUNCS DUMBELL

(DESTROY is an alias for CLOSE; SETTIMESTEP for SET\_TIME\_STEP; the spellings CUBE and DISC are lexer aliases for CUBOID and DISK; FUNCTIONS for FUNCS; and the single-b spelling DUMBELL for DUMBELL. `collisions` is deliberately **not** a keyword so the field path `system.collisions` keeps its own spelling. DEF is not a keyword either: a line starting DEF is a **line form** recognized before the ordinary grammar — see §3 and §5.9.)

Keywords are reserved *at the start of paths*, but **field names may reuse keyword spellings** — `obj0.momentum` and `system.method` work even though MOMENTUM and METHOD are keywords, because after a dot (or inside NEW { ... }) the parser accepts either an identifier or a keyword as a field name. The same holds for the scene keywords: making SHOW or IN reserved does not stop you from ever using those spellings as field names after a ..

## 2.3 Magics

A line whose first character is % is a *notebook magic* (§6.3) and is handled by the notebook itself; it never reaches the lexer.

## 3 The grammar (what the parser accepts)

The grammar in EBNF (Extended Backus–Naur Form: [x] means optional, {x} means zero-or-more repetitions, | separates alternatives):

```
line      := command | magic ;

command   := "NEW" shape [ "AS" (IDENT | STRING) ]
              [ "{" init { "," init } "}" ]
              (* AS registers a user
                 name for the object *)
              | "SET" path "=" expr
              | "GET" path
              | "DEL" NUMBER
```

```

| "LIST"
| "STEP" expr (* advance by dt *)
| "RUN" expr [ "STEPS" NUMBER ] (* advance by t, n outs *)
| "METHOD" ( "ADAMS" | "BDF" | "SPRK" IDENT [ NUMBER ] )
| "ENERGY" | "COM" | "MOMENTUM" | "ANGMOM"
| "LAPLACE" NUMBER
| "RESET" | "HELP"
| "SCENE" scenecmd (* graphical scene *)
| "COLLIDE" [ "ON" | "OFF" ] (* bare: report status *)
| "CONTACTS" (* list last contacts *)
| "LET" IDENT "=" expr (* session variable *)
| "FUNCS" (* list user functions *)
| "SHOW" IDENT (* print a function *)
| "BOX" [ "OFF" | expr ] (* rigid bounding box:
    expr = inner side
    length; bare = status;
    OFF removes it *)
| expr ; (* bare expression *)
scenecmd := "CREATE" [ NUMBER ] (* open window [port] *)
| "CLOSE" (* aka DESTROY *)
| "TRANSLATE" term term [ term ] (* camera dx dy [dz] *)
| "ROTATE" term term (* camera dyaw dpitch *)
| "ZOOM" ( "IN" | "OUT" | term ) (* factor > 1 zooms in *)
| "HIDE" [ NUMBER | "ALL" ] (* default: ALL *)
| "SHOW" [ NUMBER | "ALL" ]
| "REFRESH" (* re-sync from state *)
| "REDRAW" (* re-send full scene *)
| "START" | "STOP" | "PAUSE" | "REVERSE"
| "RESET" (* re-initialize: all
    values and the time
    return to initial;
    START re-starts *)
| "SET_TIME_STEP" term (* args are term-level:
    -5 is negative five;
    parenthesize sums *)
| "STATUS" | "EVENTS" ;
shape := "POINT" | "SPHERE" | "CUBOID" | "TORUS" | "DISK" | "CYLINDER"
| "DUMBELL" ; (* two solid spheres +
    a rigid rod, one
    rigid body *)

(* User-defined functions are a LINE FORM handled before this
    grammar: DEF name(param [= default], ...) { body } -- the body is
    newline/-separated commands using the parameters as variables;
    each body line must itself satisfy this grammar. Invocation uses
    the ordinary call syntax name(arg, ...); trailing parameters
    take their defaults. *)
init := IDENT "=" expr ;
path := IDENT { "." IDENT } ; (* objN.field[.x|y|z|w],
    system.field,
    contactK.field,
    name.field for
    AS-registered names *)
expr := sum { ("<" | "<=" | ">" | ">=" | "==" | "!=") sum } ;

```

```

(* comparisons yield 1 or 0, LOWEST
   precedence; 1/0 makes
   '(x > a) * (x < b)' an indicator
   function -- how a piecewise
   potential is written *)
sum      := term { ("+" | "-") term } ;
term     := unary { ("*" | "/" ) unary } ;
unary    := "-" unary | atom ;
atom     := NUMBER | STRING | "[" expr { "," expr } "]" | "(" expr ")"
          | IDENT "(" [ expr { "," expr } ] ")" (* builtin or user
                                                  function call *)
          | path
          | IDENT ; (* parameter / LET var *)

```

Operator precedence is the usual one: `*` and `/` bind tighter than `+` and `-`; unary minus binds tightest; parentheses override everything.

**One subtlety worth learning early:** GET takes *only a path*. To do arithmetic with a field, use a *bare expression*:

```

get obj0.position.x - 5      # ERROR -- GET wants just a path
obj0.position.x - 5          # OK -- bare expression

```

**A second subtlety — SCENE arguments are *terms*, not full expressions.** The scene commands take several numbers in a row with no commas between them. If those numbers were parsed as full expressions, `scene rotate 15 -5` would be read as *one* argument  $15 - 5 = 10$  (whitespace never matters to the lexer) and the command would then be missing its second argument. So scene arguments stop one level lower in the grammar, at `term`: `-5` is negative five, `2*2` and `1/3` still work, but a sum or difference must be parenthesized:

```

scene rotate 15 -5           # yaw +15 deg, pitch -5 deg (two arguments)
scene zoom (1 + 0.5)         # a sum needs parentheses
scene translate 2*2 0 1/2     # * and / are fine unparenthesized

```

**A third subtlety — bare identifiers compile and resolve at execution.** An `IDENT` alone is an ordinary atom: the constants `pi` and `tau`, a function parameter, or a `LET` variable (§5.9). The name is looked up when the line *runs*, not when it parses, so a function body may use parameters freely; a name that is bound to nothing fails at execution with an error that lists the three ways to bind one (`LET`, a parameter, or a registered object's `name.field` path). A call `IDENT(...)` is a builtin (`dot`, `cross`, `norm`, `normalize`, `sqrt`, `abs`, `sin`, `cos`, `exp`, `log`) when the name matches one, and a user-defined function otherwise.

## 4 The type system (what values exist)

Every expression evaluates to one of these *values*:

| type       | written as                | notes                                                                                                                                                           |
|------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| number     | 2, -0.5, 1e-3             | 64-bit float                                                                                                                                                    |
| vec3       | [1, 2, 3]                 | exactly 3 numeric entries                                                                                                                                       |
| quaternion | [w, x, y, z]              | exactly 4 numeric entries, <b>w first</b> ; assigned to <b>orientation</b> it is renormalized to unit length                                                    |
| mat3       | [[a,b,c],[d,e,f],[g,h,i]] | 3 rows of 3 numbers                                                                                                                                             |
| string     | "ball"                    | a double-quoted literal (§2.1) — names objects ( <b>NEW ... AS</b> ) and feeds user functions; also produced as results like <code>obj0</code> or run summaries |

Bracket literals are *shape-directed*: 3 numbers make a `vec3`, 4 numbers make a quaternion, 3 `vec3`s make a `mat3`.

Arithmetic is *type-checked*:

| operation | allowed                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------|
| +, -      | number±number, vec3±vec3, quat±quat                                                                      |
| *         | number·number, number·vec3, number·quat, number·mat3, mat3·vec3, mat3·mat3, quat·quat (Hamilton product) |
| /         | number/number, vec3/number                                                                               |
| vec3*vec3 | <b>forbidden</b> — the error tells you to use <code>dot()</code> or <code>cross()</code>                 |

Builtins: `dot(a,b)`, `cross(a,b)`, `norm(v)` (length; also quaternions and numbers), `normalize(v)`, and scalar `sqrt abs sin cos exp log`. Constants: `pi`, `tau` ( $= 2\pi$ ).

## 4.1 Special functions

The same call syntax reaches the `special_functions` library. Nothing about the grammar changes to accommodate them — the production `IDENT "(" [expr {"", " expr"}] ")"` already admits any builtin — so adding a function is a *registration*, not a grammar change.

**Orders must be whole numbers.** Where an argument is an integer order (the  $n$ ,  $\ell$ ,  $m$  below), a fractional value is an error rather than being quietly truncated:

```
In[1]:= hermite_h(2.5, 1)
Err[1]: hermite_h(): argument 1 must be a whole number (an integer order), got 2.5
```

That refusal is deliberate. Truncating to `hermite_h(2, 1)` would return a confident, wrong number, and you would have no way to notice.

| family                                      | functions                                                                                                                                                                     |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| spherical Bessel                            | sph_j(n,x), sph_y(n,x), sph_j_prime(n,x),<br>sph_y_prime(n,x)                                                                                                                 |
| Legendre                                    | legendre_p(n,x), legendre_p_prime(n,x),<br>assoc_legendre_p(l,m,x),<br>norm_assoc_legendre_p(l,m,x)                                                                           |
| spherical harmonics                         | sph_harm(l,m,theta,phi) → [re, im],<br>sph_harm_real(l,m,theta,phi)                                                                                                           |
| orthogonal polynomials                      | hermite_h(n,x), hermite_he(n,x), laguerre_l(n,x),<br>laguerre_l_assoc(n,alpha,x), chebyshev_t(n,x),<br>chebyshev_u(n,x), gegenbauer_c(n,alpha,x),<br>jacobi_p(n,alpha,beta,x) |
| cylindrical Bessel                          | bessel_j(n,x), bessel_j_array(n_max,x) → list                                                                                                                                 |
| cylindrical Bessel, <b>complex argument</b> | bessel_j_z(n,z), bessel_i_z(n,z),<br>bessel_y_z(n,z), bessel_k_z(n,z)                                                                                                         |
| cylindrical Bessel, <b>any real order</b>   | bessel_j_nu(nu,z), bessel_i_nu(nu,z),<br>bessel_y_nu(nu,z), bessel_k_nu(nu,z)                                                                                                 |
| Hankel (travelling waves)                   | hankel_h1_z(n,z), hankel_h2_z(n,z),<br>hankel_h1_nu(nu,z), hankel_h2_nu(nu,z)                                                                                                 |
| Hankel derivatives                          | hankel_h1_prime_z(n,z), hankel_h2_prime_z(n,z),<br>hankel_h1_prime_nu(nu,z),<br>hankel_h2_prime_nu(nu,z)                                                                      |
| spherical Hankel                            | sph_hankel_h1(n,x), sph_hankel_h2(n,x),<br>sph_hankel_h1_prime(n,x),<br>sph_hankel_h2_prime(n,x)                                                                              |
| scaled forms                                | bessel_j_scaled(nu,z), bessel_y_scaled(nu,z),<br>bessel_i_scaled(nu,z), bessel_k_scaled(nu,z),<br>hankel_h1_scaled(nu,z), hankel_h2_scaled(nu,z)                              |
| gamma, complex argument                     | gamma_z(z), ln_gamma_z(z), rgamma_z(z)                                                                                                                                        |
| Airy, complex argument                      | airy_z(z) → [Ai, Ai', Bi, Bi']                                                                                                                                                |
| quadrature                                  | gauss_legendre(n) → [nodes, weights]                                                                                                                                          |
| eigenproblems                               | eigenvalues(matrix) → list, jacobi_eigen(matrix)<br>→ [values, vectors]                                                                                                       |
| angular momentum                            | wigner_3j(j1,j2,j3,m1,m2,m3),<br>wigner_6j(j1,j2,j3,j4,j5,j6),<br>wigner_9j(a,b,c,d,e,f,g,h,i),<br>clebsch_gordan(j1,m1,j2,m2,j3,m3)                                          |
| linear algebra                              | solve_tridiag(sub, diag, sup, rhs) →<br>list, solve_tridiag_c(...) → list,<br>solve_cyclic_tridiag_c(sub, diag, sup, rhs,<br>bl, tr) → list                                   |
| utility                                     | rel_err(a, b)                                                                                                                                                                 |

**wigner\_9j** takes its nine arguments **row by row** — it recouples four angular momenta, and is the overlap between coupling (1,2) and (3,4) first versus (1,3) and (2,4) first. It is evaluated as a single sum over 6-j symbols and vanishes when any of the six triads fails to close.

**Angular momenta may be half-integers**, so **wigner\_3j**, **wigner\_6j** and **clebsch\_gordan** take



plain numbers rather than demanding whole ones — `clebsch_gordan(0.5, 0.5, 0.5, -0.5, 1, 0)` is exactly what you want to write for two spin- $\frac{1}{2}$  particles. A coupling that violates a selection rule returns **0**, which is the mathematically correct answer, not an error; a value that is not an angular momentum at all ( $j = 0.3$ , or a negative  $j$ ) is an error.

## Complex-argument Bessel

`bessel_j_z(n, z)` and `bessel_i_z(n, z)` take and return **complex** values — they exist because the language has `Value::Complex` (§4.3). Real arguments promote.

The algorithm is the same Miller downward recurrence as the real routine, deliberately: the three-term recurrence and the normalisation  $J_0 + 2(J_2 + J_4 + \dots) = 1$  are both identities in  $z$ , real or not, the second following from the generating function at  $t = 1$ .

**Accuracy of  $J$  falls with the imaginary part**, by cancellation rather than anything about the recurrence: the terms grow like  $e^{|\text{Im } z|}$  while their sum is exactly 1, so about  $|\text{Im } z|/\ln 10$  decimal digits are lost. Measured against the generating-function identity:

|                       |            |            |            |            |           |           |
|-----------------------|------------|------------|------------|------------|-----------|-----------|
| $ \text{Im } z $      | 0          | 5          | 8          | 12         | 18        | 25        |
| relative error of $J$ | $10^{-16}$ | $10^{-14}$ | $10^{-13}$ | $10^{-11}$ | $10^{-9}$ | $10^{-6}$ |

**That law governs  $J$  alone, and an earlier version of this manual let it stand for the whole family. It does not.**  $I$  is  $J$  at right angles and obeys the mirror law, but  $Y$  comes from an ascending series and  $K$  is assembled from  $J$  and  $Y$  at imaginary argument. A series fails where a recurrence does not — on the **real** axis, which is precisely where  $J$  is at its best. There are four laws:

$$\text{relative error} \sim 10^{-16} e^L, \quad L = \begin{cases} |\text{Im } z| & \text{bessel\_j\_z} \\ |\text{Re } z| & \text{bessel\_i\_z} \\ |z| - |\text{Im } z| & \text{bessel\_y\_z} \\ \max(2|\text{Re } z|, |z|) + \text{Re } z & \text{bessel\_k\_z} \end{cases}$$

Measured against Cephes on the axis where each is at its worst:

| $x$ (real)                     | 1     | 10    | 20    | 30    | 35    |
|--------------------------------|-------|-------|-------|-------|-------|
| <code>bessel_j_z(0, x)</code>  | 1e-16 | 3e-16 | 5e-16 | 2e-15 | 7e-16 |
| <code>bessel_j_z(0, ix)</code> | 2e-16 | 1e-13 | 5e-9  | 5e-5  | 3e-3  |
| <code>bessel_i_z(0, x)</code>  | 2e-16 | 1e-13 | 5e-9  | 5e-5  | 3e-3  |
| <code>bessel_y_z(0, x)</code>  | 1e-15 | 3e-12 | 3e-8  | 2e-4  | 6e-2  |
| <code>bessel_k_z(0, x)</code>  | 7e-16 | 3e-5  | 8e8   | 5e21  | 7e27  |

**That table is now history rather than guidance.** It describes what each *route* costs, and every failure in it has since been fixed by giving the function a second route:

- `bessel_j_z` near the imaginary axis uses  $J_\nu(z) = i^\nu I_\nu(-iz)$ , putting the argument back near the real axis where  $I$  has nothing to cancel;
- `bessel_y_z` near the imaginary axis uses  $Y_n(z) = i^{n+1} I_n(w) - (2/\pi) i^{-n} K_n(w)$  with  $w = -iz$ , instead of the upward recurrence in  $n$  — the wrong direction there, because  $Y_n(iy)$  is mostly  $I_n(y)$  and  $I$  is the recessive solution of that recurrence;
- `bessel_y_z` along the real axis and `bessel_k_z` everywhere use their own  $1/z$  expansions, single series with nothing to cancel;

- `bessel_j_z` and `bessel_y_z` in the wedge either side of the **negative real axis** continue the Hankel expansions from the positive one (DLMF 10.11.3, 10.11.4). Those expansions have sectors ending at  $\arg z = \pi$ , so the cut is the one direction neither reaches directly, and  $w = -z$  puts it back where both are at their best. `bessel_k_z` needed only its sector widening: DLMF 10.40.2 is valid to  $|\arg z| < 3\pi/2$ , so the whole principal sheet was always interior to it.

The routes are chosen by comparing error estimates, so nothing changes in what you write. What changed is the answers:

|                                                  | before                   | after       |
|--------------------------------------------------|--------------------------|-------------|
| <code>bessel_y_z(0, 40)</code>                   | wrong in the first digit | 1e-15       |
| <code>bessel_k_z(0, 20)</code>                   | out by a factor of 8e8   | 2e-16       |
| <code>bessel_y_z(2, 29.4e<sup>1.6i</sup>)</code> | Wronskian 4.5e-6         | below 1e-13 |

Measured across  $|z|$  from 5 to 40 and  $\arg z$  over the upper half plane, the  $J$ - $Y$  Wronskian residual is now  $10^{-10}$  **or better and mostly below**  $10^{-15}$ .

**On the cut itself that Wronskian is the wrong instrument.** It is dominated there by the exponentially *recessive* Hankel member, so it measures the Stokes phenomenon rather than the answer. The right test is the pair of exact continuation identities

$$J_n(xe^{i\pi}) = (-1)^n J_n(x), \quad Y_n(xe^{i\pi}) = (-1)^n [Y_n(x) + 2iJ_n(x)],$$

which relate a point on the cut to one on the positive real axis, where everything is at its best. Against those, `bessel_j_z` is **exact** and `bessel_y_z` is  $10^{-14}$  out to  $|z| = 300$ ; the wedge either side is  $10^{-12}$  or better, where before it reached 1.0 at  $|z| = 60$ . The branch jump is still exactly  $4i(-1)^n J_n$ , and a test says so — widening coverage across a cut is only correct if the cut stays where it was. The  $J$  and  $I$  rows of the table agree digit for digit because they are the same measurement —  $I_n(z)$  *is*  $J_n(iz)$ .

Two of those defects were found by accident and one by a test written for something else, and in each case the elementary Wronskian settled which side was wrong. The  $K$  one is worth stating plainly: its identity  $K_n(z) = (\pi/2)i^{n+1}[J_n(iz) + iY_n(iz)]$  **cancels by construction** on the real axis — both terms carry  $I_n(x)$ , of size  $e^x$ , and what survives is  $e^{-x}$ . No ingredient was inaccurate. The identity was the wrong way to compute a recessive function.

### $Y_n$ and $K_n$ : a different method, because they need one

$Y_n$  has a **logarithmic branch point** at the origin, so no recurrence produces it from  $J_n$  alone. It comes from the ascending series (DLMF 10.8.1) for  $Y_0$  and  $Y_1$  — the one carrying the  $(2/\pi)\ln(z/2)J_n(z)$  term and digamma coefficients — followed by **upward** recurrence in  $n$ .

That direction is the opposite of  $J$ 's, deliberately:  $Y_n$  *grows* with order while  $J_n$  decays, so the stable sweep for one is the unstable sweep for the other.

$K_n$  follows by identity,  $K_n(z) = (\pi/2)i^{n+1}[J_n(iz) + iY_n(iz)]$  (DLMF 10.27.8). Both are verified by Wronskians, whose right-hand sides are elementary:

$$J_{n+1}Y_n - J_nY_{n+1} = \frac{2}{\pi z}, \quad I_nK_{n+1} + I_{n+1}K_n = \frac{1}{z}.$$

Both inherit the branch cut of  $\ln$  along the negative real axis, where they jump by  $4iJ_n$  — crossing changes  $\arg$  by  $2\pi$ , and the  $(2/\pi)\ln(z/2)J$  term turns that into  $(2/\pi)(2\pi i)J$ . Both are singular at  $z = 0$  and report an error there.

## Non-integer order

The `_z` family insists on a whole order. The `_nu` family does not: `bessel_j_nu(1.3, 2 + 1i)` is  $J_{1.3}(2 + i)$ , and half-integer orders — the ones spherical Bessel functions are made of — are the common case. Order stays **real**; complex *order* is not implemented.

Non-integer order needs no recurrence at all, because the two formulas that are hardest at integer order become the easy ones here:

$$Y_\nu(z) = \frac{J_\nu(z) \cos \nu\pi - J_{-\nu}(z)}{\sin \nu\pi}, \quad K_\nu(z) = \frac{\pi}{2} \frac{I_{-\nu}(z) - I_\nu(z)}{\sin \nu\pi}.$$

Both have  $\sin \nu\pi$  downstairs, which is exactly why they are useless at whole  $\nu$  and exactly why  $Y_n$  needed its own logarithmic series. So the whole family reduces to **one ascending series** (DLMF 10.2.2 for  $J$ , 10.25.2 for  $I$ ) evaluated at  $\pm\nu$ , and the reflections do the rest. Orders within  $10^{-9}$  of a whole number are handed to the `_z` routines — not as a convenience but because by then the reflection has already lost nine digits to cancellation. `bessel_y_nu(2, z)` gives the right answer; it is `bessel_y_z(2, z)` underneath.

The series divides by  $\Gamma(\nu + k + 1)$  using the **reciprocal** gamma, which is *zero* at the poles. That is what makes  $J_{-n}(z) = (-1)^n J_n(z)$  come out for free at whole  $n$ , and it keeps very large orders in range where  $\Gamma$  itself would overflow past about 171.

**Accuracy is governed by a different law from the `_z` family**, and it is important not to carry the `_z` advice over. The largest term of the series is of size  $e^{|z|}$ , so what is lost is the ratio of that to the answer:

$$\text{relative error} \sim 10^{-16} e^L, \quad L = \begin{cases} |z| - |\operatorname{Im} z| & \text{for } J_\nu, Y_\nu \\ |z| + \operatorname{Re} z & \text{for } I_\nu, K_\nu \end{cases}$$

| $L$            | 0     | 10    | 20   | 30   | > 35         |
|----------------|-------|-------|------|------|--------------|
| relative error | 1e-16 | 1e-12 | 1e-8 | 1e-4 | nothing left |

$J$  and  $Y$  are therefore at their **worst on the real axis** and at machine precision straight up the imaginary one — good to  $|z| = 70$  there.  $I$  and  $K$  are the mirror image: worst on the **positive** real axis, exact along the negative. On the real axis that means  $|z|$  up to about 30 for  $J$  and  $Y$  but only 15 for  $I$  and  $K$ , since  $K$  is a *difference* of two  $I$ s whose leading parts cancel — that cancellation is what  $K$  *is*, not a defect of the method.

Large **order** costs nothing: at  $\nu = 150$  the order recurrence still closes to  $10^{-13}$ , because nothing cancels once  $\nu$  exceeds  $|z|$ .

Measured, not asserted:

```
cargo run -p special_functions --release --example bessel_nu_accuracy
```

prints the full error surface against the closed forms  $J_{1/2}(z) = \sqrt{2/\pi z} \sin z$  and  $K_{1/2}(z) = \sqrt{\pi/2z} e^{-z}$ , which hold for complex  $z$  and share no code with the series. The bound  $10^{-14} e^L$

is pinned by the test suite at every point of that surface. Beyond it, use the integer-order routines where the order allows — they are Miller recurrence and reach much further along the real axis. Uniform asymptotics for large  $|z|$  at non-integer order are **not** implemented.

```
In[1]:= bessell_j_nu(0.5, 2)
Out[1]= 0.5130161365618277 + 0i
In[2]:= sqrt(2/(pi*2))*sin(2)
Out[2]= 0.5130161365618278
In[3]:= bessell_j_nu(-0.5, 2)
Out[3]= -0.2347857104062484 + 0i
In[4]:= sqrt(2/(pi*2))*cos(2)
Out[4]= -0.2347857104062485
In[5]:= bessell_j_nu(1.3, 2 + 1i)
Out[5]= 0.7246607223671551 + 0.08999135538552433i
In[6]:= bessell_y_nu(2, 1.6 + 0.9i)
Out[6]= -0.6563546517741834 + 0.4094584881356723i
In[7]:= bessell_y_z(2, 1.6 + 0.9i)
Out[7]= -0.6563546517741834 + 0.4094584881356723i
In[8]:= bessell_j_z(1.5, 2)
Err[8]: bessell_j_z(): argument 1 must be a whole number (an integer order), got 1.5
```

Out[1]/Out[2] and Out[3]/Out[4] are the two half-integer closed forms agreeing to the last digit, computed two entirely different ways. Out[6] and Out[7] are byte-identical, which is the whole-order handover doing what it claims. In[8] is the contrast: the `_z` form still refuses a fractional order rather than truncating it.

## Hankel functions — the travelling-wave pair

$J$  and  $Y$  are the *standing-wave* basis of the Bessel equation.  $H^{(1)} = J + iY$  and  $H^{(2)} = J - iY$  are the *travelling-wave* basis of the same equation, and for wave problems they are the ones you want:

$$H_\nu^{(1)}(x) \sim \sqrt{\frac{2}{\pi x}} e^{i(x - \nu\pi/2 - \pi/4)},$$

a pure  $e^{+ikr}/\sqrt{r}$  outgoing cylindrical wave under the  $e^{-i\omega t}$  convention. A scattering boundary condition is stated in terms of  $H^{(1)}$ , not in terms of  $J$  and  $Y$ . The spherical pair `sph_hankel_h1(n,x)` =  $j_n(x) + iy_n(x)$  does the same job in three dimensions; it takes a **real** argument and returns a complex value.

The `_z` forms take a whole order, the `_nu` forms any real order, and `_prime` gives the derivative, from  $C'_\nu(z) = C_{\nu-1}(z) - (\nu/z)C_\nu(z)$  (DLMF 10.6.2), which holds for every cylinder function and so needs no separate algorithm.

**Why these are entry points at all.** Each is two calls to functions the language already had, and an earlier version of this manual said so and left it there. That was wrong, for the reason the accuracy note below makes concrete: the assembly  $J + iY$  silently destroys most of its digits in half the plane, and a user writing it by hand has no way to know. A named function is where that knowledge lives.

```
In[1]:= hankel_h1_z(0, 3)
Out[1]= -0.26005195490193356 + 0.37685001001279045i
In[2]:= hankel_h1_prime_z(0, 3)
```

```

Out[2]= -0.3390589585259365 - 0.3246744247917999i
In[3]:= hankel_h1_z(1, 3)
Out[3]= 0.3390589585259365 + 0.3246744247917999i
In[4]:= abs(sph_hankel_h1(3, 800) * 800)
Out[4]= 1.0000046875439457
In[5]:= hankel_h1_z(0, 0 + 10i)
Out[5]= 0 - 0.00001131945100496523i
In[6]:= hankel_h2_z(0, 0 + 10i)
Out[6]= 5631.433256931902 + 0.00001131945100496523i
In[7]:= hankel_h1_nu(0.5, 2 - 0.5i)
Out[7]= 0.7802707009570485 + 0.4802074888838277i
In[8]:= sph_hankel_h1(0, 2.3)
Out[8]= 0.32421965746813924 + 0.2896852266434018i
In[9]:= sph_hankel_h2(0, 2.3)
Out[9]= 0.32421965746813924 - 0.2896852266434018i

```

Out[2] and Out[3] are exact negatives, which is  $H_0^{(1)'} = -H_1^{(1)}$ . Out[4] is the outgoing-wave property  $|x h_n^{(1)}(x)| \rightarrow 1$ , and the gap from 1 is  $4.6875 \times 10^{-6}$ , which is  $n(n+1)/(4x^2)$  for  $n = 3$ ,  $x = 800$  to every printed digit. Out[8] and Out[9] are conjugates, as they must be for real  $x$ .

**Out[5] and Out[6] are the accuracy warning in one pair of lines.** Both come from the same  $J_0(10i)$  and  $Y_0(10i)$ , each about 2800 in magnitude.  $H^{(2)}$  comes out as 5631;  $H^{(1)}$  comes out as  $1.13 \times 10^{-5}$ , which is what is left after those two large numbers cancel. Above the real axis  $H^{(1)}$  is the small difference of large quantities and is only as accurate as that subtraction allows:

$$\text{rel. error of } H^{(1)} \sim 10^{-16} e^{3\text{Im}z} \quad (\text{Im } z > 0), \quad H^{(2)} \text{ likewise for } \text{Im } z < 0.$$

So  $H^{(1)}$  is good to  $\text{Im } z \approx 8$ , has three digits left at 10, and is gone by 12; below the axis it is exact, and  $H^{(2)}$  mirrors all of it. **Use whichever of the two is on its good side.** Switching between the `_z` and `_nu` forms does not help — measured, they agree to within a factor of 1.5, because at whole order `_nu` hands  $Y$  to `_z` and the two share the dominant ingredient. Only a scaled formulation (returning  $e^{-iz}H^{(1)}$  and letting the caller supply the exponential, as AMOS does) removes the problem, and that is **not** implemented.

That  $e^{3\text{Im}z}$  is the same law `bessel_k_z` obeys on the real axis, which is no coincidence:  $K_\nu(y) = (\pi/2)i^{\nu+1}H_\nu^{(1)}(iy)$  (DLMF 10.27.8), so they are literally the same computation.

The spherical pair has none of this trouble —  $j_n$  and  $y_n$  are recurrences on the real line, and `sph_hankel_h1` is accurate wherever they are.

Measured, not asserted:

```
cargo run -p special_functions --release --example hankel_accuracy
```

## Scaled forms — the exponential factored out

The three sections above each end in the same place: a small quantity computed from large ones, losing digits it cannot get back.  $Y_0(40)$  is wrong in its first digit,  $K_0(15)$  is worthless,  $H^{(1)}$  above the real axis is gone by  $\text{Im } z = 12$ . Each said the remedy was “a scaled formulation, which is not implemented”. It is now.

| function               | returns                               |
|------------------------|---------------------------------------|
| bessel_j_scaled(nu,z)  | $e^{- \operatorname{Im} z } J_\nu(z)$ |
| bessel_y_scaled(nu,z)  | $e^{- \operatorname{Im} z } Y_\nu(z)$ |
| bessel_i_scaled(nu,z)  | $e^{- \operatorname{Re} z } I_\nu(z)$ |
| bessel_k_scaled(nu,z)  | $e^z K_\nu(z)$                        |
| hankel_h1_scaled(nu,z) | $e^{-iz} H_\nu^{(1)}(z)$              |
| hankel_h2_scaled(nu,z) | $e^{iz} H_\nu^{(2)}(z)$               |

**Multiplying by the exponential afterwards would fix nothing** — the digits are gone before the multiplication. What makes these work is a different algorithm: the asymptotic expansions of DLMF 10.17 and 10.40, plain series in  $1/z$  with leading term 1 and **no cancellation anywhere in them**:

$$e^{-iz} H_\nu^{(1)}(z) \sim \sqrt{\frac{2}{\pi z}} e^{-i(\nu\pi/2 + \pi/4)} S(i), \quad e^z K_\nu(z) \sim \sqrt{\frac{\pi}{2z}} S(1),$$

$$S(c) = \sum_k \frac{c^k a_k(\nu)}{z^k}, \quad a_0 = 1, \quad a_k = a_{k-1} \frac{4\nu^2 - (2k-1)^2}{8k}.$$

$J$  and  $Y$  then come from the Hankel pair without the growing exponential ever being formed, because  $e^{\pm iz - |\operatorname{Im} z|}$  has modulus at most 1 by construction.

```
In[1]:= bessel_y_scaled(0, 40)
Out[1]= 0.12593641705826092 + 0i
In[2]:= bessel_y_z(0, 40)
Out[2]= 0.09242859229058376 + 0i
In[3]:= bessel_k_scaled(0, 2000)
Out[3]= 0.02802320501460432 + 0i
In[4]:= sqrt(pi/4000)
Out[4]= 0.028024956081989644
In[5]:= bessel_i_scaled(0, 1000)
Out[5]= 0.012617240455891252 + 0i
In[6]:= hankel_h1_scaled(0.5, 3 + 700i)
Out[6]= -0.02127852045069675 - 0.02136990952385759i
In[7]:= bessel_k_scaled(40, 25)
Out[7]= 213915362812.9533 + 0i
In[8]:= bessel_y_scaled(0, 5000)
Out[8]= -0.009116740769643965 + -0i
In[9]:= bessel_k_scaled(0.5, 1e6)
Out[9]= 0.0012533141373154998 + 0i
In[10]:= sqrt(pi/2e6)
Out[10]= 0.0012533141373155003
```

On the real axis the  $J/Y$  scaling is  $e^0 = 1$ , so Out[1] is  $Y_0(40)$  and Out[2] is the same number computed the old way. The true value is 0.12593641705826097: the scaled form has fifteen correct digits, the unscaled one none.

Out[3] and Out[4] are  $e^x K_0(2000)$  and its leading asymptotic  $\sqrt{\pi/2x}$ , agreeing to four digits with the gap equal to  $-1/(8x)$  to the digit. **Unscaled, that value does not exist:**  $K_0(2000)$  is about  $10^{-870}$ , below the smallest f64. Out[5] is the same story upward,  $I_0(1000) \approx e^{1000}$  being above the largest. Out[6] is  $H^{(1)}$  seven hundred nepers above the real axis, where  $J$  and  $Y$  are each about  $e^{700}$ . Out[9]/Out[10] are the exact half-integer form  $e^z K_{1/2}(z) = \sqrt{\pi/2z}$  at  $z = 10^6$ .

**Order is handled by recurrence.**  $K$ ,  $Y$ ,  $H^{(1)}$  and  $H^{(2)}$  all grow with order, so upward recurrence in  $\nu$  is stable; the expansion is used at a base order below 1 and stepped up. `Out[7]` is order 40, where the vendored Cephes `kn` overflows outright.

**Large order is a separate expansion.** A series in  $1/z$  at fixed order cannot reach  $z$  below  $\nu$ , where  $J$  is exponentially small and was being built as the difference of two exponentially large Hankel values — at  $\nu = 400.5$ ,  $z = 240$  that gave an answer wrong by a factor of  $5 \times 10^{89}$ . The remedy is a series in  $1/\nu$ : the **Debye polynomials** and the uniform expansions of DLMF 10.19 and 10.41,

$$U_0(p) = 1, \quad U_{k+1}(p) = \frac{1}{2}p^2(1-p^2)U'_k(p) + \frac{1}{8} \int_0^p (1-5t^2)U_k(t) dt,$$

which produce the small number directly. They are chosen automatically by comparing error estimates, so nothing in the language changes; what changes is that the answers are now right. Measured against Cephes over orders 10 to 1000 and  $z/\nu$  from 0.1 to 2, every value is within  $10^{-9}$  and most within  $10^{-14}$  — and at  $\nu = 400.5$  it is *Cephes* that is looser, by  $1.4 \times 10^{-9}$ , which the elementary  $J$ - $Y$  Wronskian adjudicates.

**When no method reaches a point these return an error** naming both estimates. Since the large-order expansions arrived, most such points instead fail for a different and better-stated reason: *the value is outside f64*. For  $z$  well below  $\nu$ ,  $J$  is smaller than the smallest double and  $Y$  larger than the largest, and the error says so and quotes the logarithm.

**The turning point has its own expansion.** Olver's uniform Airy-type expansion (DLMF 10.20) replaces the elementary prefactor with an Airy function, the exact local model of where the equation stops oscillating:

$$J_\nu(\nu x) \sim \left(\frac{4\zeta}{1-x^2}\right)^{1/4} \left[ \frac{\text{Ai}(\nu^{2/3}\zeta)}{\nu^{1/3}} \sum_k \frac{A_k(\zeta)}{\nu^{2k}} + \frac{\text{Ai}'(\nu^{2/3}\zeta)}{\nu^{5/3}} \sum_k \frac{B_k(\zeta)}{\nu^{2k}} \right].$$

Its coefficients are built from the same Debye polynomials, and every term in them is singular at  $\zeta = 0$  with the singularities cancelling exactly — which is why near the turning point they come from Taylor series in  $1-x$  generated at 70 digits instead. Two of the resulting constants are known independently and match:  $A_1(0) = -1/225$  and  $\zeta'(1) = -2^{1/3}$ .

An earlier version of this manual argued that 10.20 was unnecessary, because across  $z/\nu$  from 0.95 to 1.1 the other routes already gave  $10^{-14}$ . **That measurement was too coarse and the conclusion was wrong.** At  $z/\nu = 0.85$  and  $0.90$  — not sampled — the error reached  $1.4 \times 10^{-9}$ . With 10.20 in the mix the whole band is  $10^{-14}$ . Complex *order* remains absent.

Measured, not asserted:

```
cargo run -p special_functions --release --example bessel_scaled_accuracy
```

## Complex order

Every `_nu` form above takes a **complex** order as well as a real one. `bessel_j_nu(1 + 2i, 3)` is  $J_{1+2i}(3)$ ; a real order reaches exactly the routine it always did, so nothing that worked before changes.

This was the last entry on the list of things chapter 10 did not cover, and the one no expansion could have closed. The obstacle was not an algorithm — the ascending series is the same — it

was that  $1/\Gamma(\nu + k + 1)$  had no meaning here for complex  $\nu$ . So the stage is really a **complex gamma**, and the Bessel functions follow.

|                            |                                                        |
|----------------------------|--------------------------------------------------------|
| <code>gamma_z(z)</code>    | $\Gamma$ at complex argument                           |
| <code>ln_gamma_z(z)</code> | still defined where $\Gamma$ has left <code>f64</code> |
| <code>rgamma_z(z)</code>   | $1/\Gamma$ , <b>entire</b> — exactly zero at the poles |

`gamma_z` is Stirling with argument shifting, not Lanczos. Lanczos is a little faster and is rejected for the reason that has shaped this crate: its coefficients are a *table*, and the tables in circulation are most often reproduced from *Numerical Recipes*, whose licence this project will not inherit. Stirling needs no table — only the Bernoulli numbers, defined by a recurrence the crate states and a test re-derives.

```
In[1]:= gamma_z(0.5)
Out[1]= 1.7724538509055292 + 0i
In[2]:= sqrt(pi)
Out[2]= 1.7724538509055159
In[3]:= gamma_z(1 + 1i)
Out[3]= 0.4980156681183574 - 0.15494982830181092i
In[4]:= rgamma_z(-3)
Out[4]= 0 + 0i
In[5]:= ln_gamma_z(200)
Out[5]= 857.9336698258575 + 0i
In[6]:= bessej_nu(1 + 2i, 3)
Out[6]= 2.616967138257939 + 0.5245621513264704i
In[7]:= bessej_nu(0.5, 2)
Out[7]= 0.5130161365618277 + 0i
In[8]:= bessek_nu(1i, 2)
Out[8]= 0.09238545989039124 + 0i
In[9]:= bessek_nu(-1i, 2)
Out[9]= 0.09238545989039124 + 0i
```

`Out[1]` and `Out[2]` differ in the fourteenth digit, which is the method's honest accuracy: reaching the Stirling regime from  $z = 0.5$  takes fourteen shifts and each rounds. `Out[4]` is the reciprocal being *entire*. `Out[5]` is a value `gamma_z(200)` refuses,  $\Gamma(200)$  being about  $10^{372}$ .

**Out[8] and Out[9] are the result worth looking at.**  $K_{iy}(x)$  is real for real  $y$  and real positive  $x$  — the Macdonald function of imaginary order, which is why it appears as an eigenfunction on the half line. Nothing in the implementation arranges that:  $K$  is built from two  $I$ s at  $\pm\nu$  divided by  $\sin\nu\pi$ , all three thoroughly complex. It comes out real, and even in the order, because the mathematics says so.

**Two things change once the order is complex**, both because it now sits in an exponent.  $(z/2)^\nu = e^{\nu \ln(z/2)}$ , so its *modulus* depends on  $\arg z$  as well as  $|z|$  — a branch choice is no longer a phase convention. And  $\sin\nu\pi$  in the reflections grows like  $e^{\pi|\operatorname{Im}\nu|}$ , so they get **better** conditioned off the real axis: a whole-numbered real part is not a special case unless the imaginary part is small too. The accuracy law gains one term for the same reason:

$$\text{relative error} \sim 10^{-16} \exp(|z| - |\operatorname{Im} z| + \operatorname{Im} \nu \cdot \arg z),$$

so complex order is free on the positive real axis and costs  $\operatorname{Im} \nu \cdot \arg z$  elsewhere.

**Beyond the series.** The  $1/z$  asymptotics (DLMF 10.17, 10.40) are expansions at *fixed* order



in which the order enters only as  $\mu = 4\nu^2$  — polynomially — so nothing in them assumes a real order and they extend unchanged. The uniform expansions of DLMF 10.41 for  $I$  and  $K$  extend in the sector  $|\arg \nu| < \pi/2$  (DLMF 10.41.5). Both are offered alongside the series and chosen by comparing error estimates, so the reach runs from  $|z| \approx 25$  out past  $|z| = 600$ .

Two things the estimates could not see had to be measured. The  $1/z$  expansion needs  $|4\nu^2|$  small compared with  $|z|$ , not merely  $|z|$  large — at  $\nu = 0.5 + 5i$  and  $|z| = 22$  the actual error was **2163 times** the truncation estimate — so that is checked as a validity condition rather than folded into a tolerance. And the term DLMF 10.40.1 drops from  $I$  is not of relative size  $e^{-2\operatorname{Re} z}$  but  $e^{\pi|\operatorname{Im} \nu| - 2\operatorname{Re} z}$ : **complex order makes it bigger**, by a factor this crate had no reason to carry until now.

**The turning point is covered too.** Olver’s Airy-type expansion (DLMF 10.20) now runs at complex order, using the complex Airy functions of `airy_z`. It needed no new mathematics: near  $x = 1$  every ingredient —  $\zeta$ , the prefactor, and the coefficients  $A_k, B_k$  — is already a Taylor series in  $w = 1 - x$ , generated at 70 digits, and a Taylor series does not care whether its variable is real. What did *not* continue to complex  $x$  are the closed forms outside that neighbourhood, where 10.20 is the wrong tool anyway.

**The Debye region is covered too**, on both sides of the turning point and at complex order as well as real. With  $t = \sqrt{1 - x^2}$ ,  $\alpha = \ln((1 + t)/x)$ ,  $q = 1/t$  and  $x = z/\nu$ ,

$$F_{\pm}(\nu, x) = \frac{e^{\pm\nu(t-\alpha)}}{\sqrt{2\pi\nu t}} \sum_k \frac{(\pm 1)^k U_k(q)}{\nu^k}$$

reads two ways. For  $|x| < 1$  these are DLMF 10.19.3/4 directly,  $J = F_+$  and  $Y = -2F_-$ . For  $|x| > 1$  — the **oscillatory** region, where  $t$  turns imaginary — they continue into the Hankel functions instead,  $H^{(1)} = 2F_+$  and  $H^{(2)} = 2iF_-$ , so  $J = F_+ + iF_-$  and  $Y = -iF_+ - F_-$ . That flip is the Stokes phenomenon, and **the constants were identified by experiment rather than transcribed**: continuing  $F_+$  past  $x = 1$  and dividing by each of  $J, Y, H^{(1)}, H^{(2)}$  in turn showed  $F_+/H^{(1)} = \frac{1}{2}$ .

That band had no method at all, at real order as well as complex. `bessel_y_z(20, 60)` used to return  $10^8$ ; it is now correct to  $2 \times 10^{-14}$ , and across orders 8 to 150 and  $z/\nu$  from 2 to 30 the whole band is  $10^{-14}$  or better.

What remains is a sliver:  $|\nu|$  between about 4 and 8 with  $|z|$  a few times larger, where the  $1/z$  route is refused for  $|4\nu^2|$  and the Debye one for being a  $1/\nu$  series at too small an order. Both refusals are deliberate and measured. There is also a narrow ridge near  $z/\nu \approx 1.3$  where  $Y$  reaches only about  $10^{-7}$  at moderate order.

## Airy functions, complex argument

`airy_z(z)` returns all four at once — `[Ai, Ai', Bi, Bi']` — because the routine computes them together and a caller who wants a Wronskian or a turning-point boundary condition wants all four.

```
In[1]:= airy_z(0)
Out[1]= [0.3550280538878172 + 0i, -0.2588194037928068 + 0i,
         0.6149266274460007 + 0i, 0.4482883573538264 + 0i]
In[2]:= airy_z(2 - 3i)
Out[2]= [0.008104457809530868 - 0.131178382604566i,
         0.09665817903311252 + 0.23198718538548577i,
         -0.3963682550403918 + 0.5697309129559497i,
```

Out[1] is the closed form at the origin:  $\text{Ai}(0) = 3^{-2/3}/\Gamma(2/3)$ ,  $\text{Ai}'(0) = -3^{-1/3}/\Gamma(1/3)$  and  $\text{Bi}(0) = \sqrt{3} \text{Ai}(0)$ , the last of which you can read straight off the numbers.

**Three regimes, one connection formula.** The ascending series (DLMF 9.4) up to  $|z| \approx 6$ ; the asymptotic expansions (DLMF 9.7.5, 9.7.6) in  $\zeta = \frac{2}{3}z^{3/2}$  for  $|\arg z| \leq 2\pi/3$ ; and nearer the negative real axis the connection  $\text{Ai}(z) + \omega \text{Ai}(\omega z) + \omega^2 \text{Ai}(\omega^2 z) = 0$  with  $\omega = e^{2\pi i/3}$ , which rotates  $\arg z \sim \pi$  to  $\arg \sim \pm\pi/3$  where the expansion is at its best — the same trick the Bessel functions use for their own cut, and it works for the same reason.

Bi needs no expansion of its own past the series: its asymptotic is valid only for  $|\arg z| < \pi/3$ , and  $\text{Bi}(z) = e^{i\pi/6} \text{Ai}(ze^{2\pi i/3}) + e^{-i\pi/6} \text{Ai}(ze^{-2\pi i/3})$  gives it everywhere.

**How it is checked.** The Wronskian  $\text{Ai Bi}' - \text{Ai}' \text{Bi} = 1/\pi$  is exact, elementary, and free of both the argument and any transcendental function on the right. Across  $|z|$  from 0.1 to 200 and the full range of  $\arg z$  the residual is  $10^{-11}$  **or better**, except in one band: between  $|z| = 3$  and 10 it is  $10^{-9}$  — the crossover, where Ai is exponentially recessive. The test states that band separately rather than loosening the whole bound to accommodate it. Also checked: the connection formula away from where it is used, conjugation symmetry, the defining equation  $\text{Ai}'' = z \text{Ai}$  by finite difference, and Cephes on the real axis.

Past  $|z| \approx 90$  off the real axis the *dominant* solution leaves **f64** — it grows like  $e^{2|z|^{3/2}/3}$  — and **airy\_z** says so rather than returning an infinity.

**A wrinkle worth knowing about lists.** The bracket literal is overloaded: **[a,b,c]** is a *vector*, **[a,b,c,d]** is a *quaternion*, and any other length is a list. That is convenient for physics and would be a nuisance here, so all three shapes are accepted anywhere a numeric list is expected — a  $4 \times 4$  matrix whose rows are 4-element brackets works exactly as you would expect, and **norm\_assoc\_legendre\_p** never lectures you about quaternions. A  $3 \times 3$  matrix value is likewise accepted directly wherever a matrix argument is wanted.

## 4.2 Comparisons

**<, <=, >, >=, ==, !=** compare two numbers and yield **1 for true, 0 for false**. There is no boolean type, and that is the point:

```
In[1]:= 3 < 5
Out[1]= 1
In[2]:= (0.5 > 0) * (0.5 < 1)
Out[2]= 1
In[3]:= (2 > 0) * (2 < 1)
Out[3]= 0
```

**(x > a) \* (x < b)** is an **indicator function** for the interval  $(a, b)$ . Multiply it by a height and you have a rectangular barrier; add two and you have a double barrier. This is how a piecewise potential is written — see §5.10 and Example 17.

Comparisons sit at the **lowest** precedence, below + and −, so **x + 1 > 2** groups as **(x + 1) > 2**. They are left associative, so **a < b < c** means **(a < b) < c**: legal, and almost certainly not what you meant.

= is still assignment; == is equality. Ordering is defined for numbers only; == and != also accept two vectors or two quaternions, because asking whether two positions coincide is reasonable while ordering them is not.

**NaN compares false to everything, including itself**, so `x != x` is the idiomatic NaN test:

```
In[4] := 0/0 != 0/0
Out[4] = 1
```

That is not special-cased — it falls out of IEEE-754.

### 4.3 Complex numbers

A number with an `i` suffix is imaginary, so `2 + 3i` needs no complex literal syntax of its own — it is ordinary addition of a real and an imaginary:

```
In[1] := 3i
Out[1] = 0 + 3i
In[2] := 2 + 3i
Out[2] = 2 + 3i
In[3] := (2 + 3i) * (2 - 3i)
Out[3] = 13 + 0i
In[4] := 1 / (0 + 1i)
Out[4] = 0 - 1i
```

`+`, `-`, `*` and `/` all accept complex operands, and reals promote automatically. Note `In[3]`: the result stays typed as complex and displays `13 + 0i` rather than collapsing to `13`. That is deliberate — the type you get out should not depend on whether a particular cancellation happened to be exact.

The suffix only binds when the `i` is not followed by another identifier character, so `2intercept` still lexes exactly as it did before complex numbers existed.

This is what makes the complex Crank–Nicolson solvers reachable. `solve_tridiag_c` and `solve_cyclic_tridiag` take the same arguments as the real solver and accept a mix — a real band with a complex diagonal and a real right-hand side is fine, since reals promote:

```
In[5] := solve_tridiag_c([0, 1, 1], [1i, 1i, 1i], [1, 1, 0], [1, 0, 0])
Out[5] = [0 - 0.6666666666666666i, 0.3333333333333337 + 0i, 0 + 0.3333333333333333i]
```

Over the machine bridge (JupyterLab), a complex value is reported as the two-element array `[re, im]`, matching how `sph_harm` reports its value — JSON has no complex type.

## 5 Command semantics

### 5.1 NEW — create an object

```
NEW POINT    { field = expr, ... }
NEW SPHERE   { field = expr, ... }
NEW CUBOID   { field = expr, ... }      (alias: NEW CUBE)
NEW TORUS    { field = expr, ... }
NEW DISK     { field = expr, ... }      (alias: NEW DISC)
NEW CYLINDER { field = expr, ... }
```

|                                                                    |
|--------------------------------------------------------------------|
| NEW DUMBELL { field = expr, ... } (alias: NEW DUMBELL)             |
| NEW <shape> AS <name> { field = expr, ... } (register a user name) |

Creates one *physical object* and prints its handle (obj0, obj1, ..., numbered by position). Defaults: mass 1, at the origin, at rest, no charge; sphere radius 1; cuboid half-extents [1,1,1]; torus ring\_radius 1, tube\_radius 0.25; disk radius 1; cylinder radius 0.5, half\_height 1; dumbbell m1 = 1, m2 = 1, m\_rod = 0.5, r1 = 0.25, r2 = 0.25, rod\_radius = 0.1, length = 1 (total mass 2.5).

Initializer fields: mass, charge, position, velocity, momentum, orientation, angular\_velocity, angular\_momentum, radius (spheres, disks, cylinders), half\_extents (cuboids), ring\_radius, tube\_radius (tori — or the inner\_radius + outer\_radius pair, see below), height, half\_height (cylinders; HEIGHT is the full height = 2·half\_height), m1, m2, m\_rod, r1, r2, rod\_radius, length (dumbbells, see below), inertia\_tensor, inverse\_inertia\_tensor, magnetic\_moment\_tensor, force, torque, id, inverse\_mass.

A torus can be sized two ways: directly (ring\_radius  $c$  = the radius of the circle traced by the tube's center, tube\_radius  $a$  = the tube's own radius) or by the **inner\_radius + outer\_radius pair** (the hole's radius and the outermost radius: inner =  $c - a$ , outer =  $c + a$ ). Inside a NEW initializer list the torus geometry is **deferred**: the four radius fields are collected and resolved *once*, at the end of the list — ring\_radius/tube\_radius are applied first, then inner\_radius/outer\_radius override the derived pair — and the result is validated once, against the *final* values (a bad pair fails with `torus needs 0 <= inner < outer` (got inner = 2, outer = 1)). Giving both members of the pair is therefore **genuinely order-independent**: { inner\_radius = 1, outer\_radius = 2 } yields ring 1.5, tube 0.5 whichever comes first, and a pair that shrinks the default torus — { outer\_radius = 0.5, inner\_radius = 0.2 } — works in either order too (checking each write against the half-updated default would have refused one of the two orders). inner\_radius = 0 is legal — that is the **horn torus**, whose tube touches the axis — on NEW and on SET alike.

**The dumbbell** is ONE rigid body: two solid spheres joined by a solid rod. Its seven part fields are m1, m2, m\_rod (the two sphere masses and the rod mass), r1, r2 (the sphere radii), rod\_radius (alias rod\_r) and length (alias len — the distance between the sphere centers). Like the torus radius pair, the part fields are **deferred**: they are collected during the initializer list and resolved *once* at its end, so they are order-independent and validated once against the final values. The constructor places the local origin at the composite **center of mass** — the sphere centers sit at  $z_1 = -(m_2 + m_{\text{rod}}/2) L/M$  and  $z_2 = +(m_1 + m_{\text{rod}}/2) L/M$  on the body  $z$ -axis, which makes  $m_1 z_1 + m_2 z_2 + m_{\text{rod}}(z_1 + z_2)/2 = 0$  an identity — so **position** is the COM, exactly as for every other shape. The inertia tensor is the exact composite (solid-sphere  $\frac{2}{5}mr^2$  terms with parallel-axis shifts, plus the rod), and the surface is the exact union of the three parts. It is the simulator's first non-centrally-symmetric shape; collisions against every other shape decompose exactly over its parts (see `collision_detection.pdf`). The part fields remain readable *and writable* after creation (§5.2).

**AS registers a user name for the object.** The name is a string literal (NEW SPHERE AS "ball") or a bare identifier (NEW SPHERE AS ball); a bare identifier is first resolved against a function parameter or LET variable holding a string — so a function can name the object it creates from its argument (§5.9, Example 14) — and otherwise the identifier's own spelling is the name. The reply shows both handles: obj0 as ball. Named paths then work everywhere a positional path does —

`ball.mass`, `dumbell0.position.x`, in SET/GET and in bare expressions. Names are case-insensitive identifiers; `system/sys` and the positional spellings are refused (`'system'` is reserved, `'obj7'` is reserved for positional paths), and so are duplicates (the name `'d'` already refers to `obj0` -- DEL it or pick another name). The registry follows every renumbering: DEL removes the deleted object's name and shifts the names of higher-numbered objects down with their objects, and the BOX commands do the same when walls come and go.

Four guarantees make initializers forgiving:

1. **Order does not matter for velocities.** `velocity` and `angular_velocity` are applied *after* `mass` and `inertia` are final, so `{ velocity = [1,0,0], mass = 2 }` still yields momentum `[2,0,0]`.
2. **Inertia is computed for you.** For every extended shape the inertia tensor is recomputed from the final mass and shape (solid sphere  $\frac{2}{5}mr^2$ ; cuboid  $\frac{m}{3}(h_y^2 + h_z^2)$  etc.; torus  $I_z = m(c^2 + \frac{3}{4}a^2)$ ,  $I_{xy} = m(\frac{1}{2}c^2 + \frac{5}{8}a^2)$ ; disk  $I_z = \frac{1}{2}ma^2$ ,  $I_{xy} = \frac{1}{4}ma^2$ ; cylinder  $I_z = \frac{1}{2}mr^2$ ,  $I_{xy} = m(3r^2 + 4h^2)/12$  with  $h$  the half-height) — *unless* you supplied `inertia_tensor` yourself, in which case yours is kept.
3. **Coupled fields stay consistent** (see SET below).
4. **NEW is transactional.** If any initializer — or the final geometry validation — fails, the half-built object is removed before the error is reported: a failing **NEW** never leaves a ghost behind, and `system.count` and the `objN` numbering are exactly what they were before the command.

## 5.2 SET / GET — write and read any field

A *path* is `objN.field`, `system.field`, `contactK.field` (§5.7) or `name.field` for a name registered with **NEW** ... **AS** (§5.1), optionally followed by a component `.x .y .z` (vectors) or `.w .x .y .z` (quaternions). Component writes are safe read-modify-write operations through the full field's get/set pair. Every object also has six **component shorthands**: `.x .y .z` read and write the position components and `.vx .vy .vz` the velocity components — `ball.x` is `ball.position.x`, `SET ball.vx = 2` is a velocity write.

**Object fields** (R = readable, W = writable):

| field                         | type   | R/W | meaning                                                                               |
|-------------------------------|--------|-----|---------------------------------------------------------------------------------------|
| <code>id</code>               | number | RW  | user label (no effect on physics)                                                     |
| <code>mass</code>             | number | RW  | writing also updates <code>inverse_mass</code><br>( $m \leq 0 \Rightarrow 1/m := 0$ ) |
| <code>inverse_mass</code>     | number | RW  | writing back-computes <code>mass</code> ; 0 makes the body <b>static</b>              |
| <code>charge</code>           | number | RW  | Coulombs                                                                              |
| <code>position, pos</code>    | vec3   | RW  | world position                                                                        |
| <code>velocity, vel</code>    | vec3   | RW  | <i>derived</i> : reads $p/m$ , writes $p := mv$                                       |
| <code>momentum</code>         | vec3   | RW  | canonical stored linear state                                                         |
| <code>orientation</code>      | quat   | RW  | renormalized on write                                                                 |
| <code>angular_velocity</code> | vec3   | RW  | <i>derived</i> : $\omega = R I^{-1} R^T L$ ; writes $L := R I R^T \omega$             |
| <code>angular_momentum</code> | vec3   | RW  | canonical stored angular state (spin)                                                 |

|                            |        |    |                                                                                                                                                                                                                                                                                                                   |
|----------------------------|--------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| inertia_tensor             | mat3   | RW | body frame; write updates the inverse (singular $\rightarrow$ zero inverse = cannot rotate)                                                                                                                                                                                                                       |
| inverse_inertia_tensor     | mat3   | RW | write back-computes the tensor                                                                                                                                                                                                                                                                                    |
| magnetic_moment_tensor     | mat3   | RW | torque $\tau = (RMR^T)B$                                                                                                                                                                                                                                                                                          |
| radius                     | number | RW | spheres, disks, cylinders; write recomputes inertia and <b>keeps the shape family</b> (a disk stays a disk, a cylinder keeps its height); a torus is <b>refused</b> — obj6 is a torus — set ring_radius/tube_radius or inner_radius/outer_radius instead of radius — rather than silently becoming a sphere       |
| half_extents               | vec3   | RW | cuboids only; write recomputes inertia                                                                                                                                                                                                                                                                            |
| ring_radius, tube_radius   | number | RW | tori only; write recomputes inertia                                                                                                                                                                                                                                                                               |
| inner_radius, outer_radius | number | RW | tori only, derived (inner = $c - a$ , outer = $c + a$ ); writing one holds the other fixed; inner_radius accepts $\geq 0$ (0 = the horn torus; the other radii must be $> 0$ )                                                                                                                                    |
| height, half_height        | number | RW | cylinders only; height is the full height (= $2 \cdot \text{half\_height}$ ); write recomputes inertia                                                                                                                                                                                                            |
| m1, m2, m_rod              | number | RW | dumbbells only: the two sphere masses and the rod mass; writing rebuilds the body — total mass, the COM offsets and the inertia tensor all follow (position is untouched; the stored momenta are preserved — momentum-canonical, like every mass write — so the velocities rescale with the new mass and inertia) |
| r1, r2, rod_radius         | number | RW | dumbbells only: the sphere radii and the rod radius (alias rod_r); same rebuild on write                                                                                                                                                                                                                          |
| length                     | number | RW | dumbbells only: distance between the sphere centers (alias len); same rebuild on write                                                                                                                                                                                                                            |
| x, y, z                    | number | RW | every object: shorthands for position.x/.y/.z                                                                                                                                                                                                                                                                     |
| vx, vy, vz                 | number | RW | every object: shorthands for velocity.x/.y/.z                                                                                                                                                                                                                                                                     |
| boundary, shape            | string | R  | description of the shape                                                                                                                                                                                                                                                                                          |
| kinetic_energy, energy     | number | R  | $\frac{1}{2}m v ^2 + \frac{1}{2}\omega \cdot L$                                                                                                                                                                                                                                                                   |
| force                      | vec3   | RW | constant external force during STEP/RUN                                                                                                                                                                                                                                                                           |

|             |        |    |                                                                                          |
|-------------|--------|----|------------------------------------------------------------------------------------------|
| torque      | vec3   | RW | constant external torque                                                                 |
| restitution | number | RW | collision bounciness $e \in [0, 1]$ (default 1 = elastic; a pair uses $\min(e_i, e_j)$ ) |

#### System fields:

| field                    | type   | R/W | meaning                                                                                     |
|--------------------------|--------|-----|---------------------------------------------------------------------------------------------|
| g_constant, g            | number | RW  | Newton's $G$ (default 1)                                                                    |
| softening                | number | RW  | Plummer length $\varepsilon$ (default $10^{-6}$ ); forces use $(r^2 + \varepsilon^2)^{3/2}$ |
| uniform_gravity, gravity | vec3   | RW  | constant acceleration field                                                                 |
| e_field                  | vec3   | RW  | uniform $E$ ; force $qE$                                                                    |
| b_field                  | vec3   | RW  | uniform $B$ ; force $qv \times B$ , torque $(RM R^T)B$                                      |
| rtol, atol               | number | RW  | CVODE tolerances ( $10^{-10}$ , $10^{-12}$ )                                                |
| time, t                  | number | RW  | current simulation time                                                                     |
| method                   | string | R   | current integrator                                                                          |
| count, n                 | number | R   | number of objects                                                                           |
| collide                  | number | R   | 1 when collision detection is on (switch with COLLIDE ON/OFF)                               |
| contacts                 | number | R   | contacts recorded by the last STEP/RUN                                                      |
| collisions               | number | R   | running total of resolved impulses                                                          |
| restitution_threshold    | number | RW  | approach speeds below this bounce with $e = 0$ (default $10^{-3}$ )                         |
| contact_slop             | number | RW  | tolerated overlap before positional projection (default $10^{-9}$ )                         |
| box                      | number | R   | inner side length of the rigid bounding box (§5.8); 0 = none. Create/remove with BOX        |

### 5.3 STEP and RUN — integrate (always via SUNDIALS)

STEP  $dt$  advances time by  $dt$ . RUN  $t$  STEPS  $n$  advances by  $t$ , stopping at  $n$  evenly spaced output points (default 10). Both accept full expressions (run  $2 * \pi$  steps 8 is legal). The reply summarizes the run;  $|dE/E|$  is the relative change in total energy — your built-in sanity check. **All integration is performed by the pure-Rust SUNDIALS solvers**; there is no hand-rolled stepper.

### 5.4 METHOD — choose the integrator

|                          |                                                  |
|--------------------------|--------------------------------------------------|
| METHOD ADAMS             | (default; CVODE Adams-Moulton, adaptive)         |
| METHOD BDF               | (CVODE BDF -- stiff problems, fast gyration)     |
| METHOD SPRK <table> [dt] | (ARKODE symplectic, fixed step; default dt 0.01) |

Table names may be abbreviated: leapfrog\_2\_2 becomes ARKODE\_SPRK\_LEAPFROG\_2\_2. Useful tables: EULER\_1\_1, LEAPFROG\_2\_2 ( $\equiv$  classic velocity Verlet), MCLACHLAN\_2\_2/3\_3/4\_4/5\_5/6\_6, RUTH\_3\_3, YOSHIDA\_6\_8.

SPRK requires a *separable* system: point-like translation only. If anything velocity-dependent or rotational is active (a `b_field`, a magnetic tensor, an external torque, a spinning rigid body), RUN refuses with an error *naming the offending feature*.

## 5.5 Observables, bookkeeping, help

| command     | prints                                                                                                                                                |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| ENERGY      | kinetic + softened pairwise potential + uniform-field potentials                                                                                      |
| COM         | system center of mass                                                                                                                                 |
| MOMENTUM    | total linear momentum                                                                                                                                 |
| ANGMOM      | total angular momentum about the origin (orbital + spin)                                                                                              |
| LAPLACE $n$ | Laplace–Runge–Lenz vector of object $n$ about the center of mass, $k = G M_{\text{total}}$                                                            |
| LIST        | one line per object                                                                                                                                   |
| DEL $n$     | removes object $n$ ( <b>later objects renumber!</b> )                                                                                                 |
| RESET       | wipes everything (not to be confused with SCENE RESET, which re-initializes only the scene window’s playback copy — see <code>scene_info.pdf</code> ) |
| HELP        | the quick-reference card                                                                                                                              |

## 5.6 SCENE — the graphical scene window

SCENE CREATE starts a tiny web server *inside* posim (pure Rust standard library — no external dependencies) and opens a page in your web browser. That page **is** the scene window: it draws every simulator entity on a 3-D canvas, has a **toolbar** (Start, Pause, Stop, Reverse, a permanent Reset button, single-step, a dt box, zoom, view reset, grid/trails/labels toggles, help) and a **status bar** (connection light, playback mode, simulation time, dt, total energy, body count, hidden count, history depth, camera readout, frames per second). Inside the window:

| gesture                               | effect                                          |
|---------------------------------------|-------------------------------------------------|
| $\leftarrow \rightarrow$ (arrow keys) | translate the view left / right                 |
| $\uparrow \downarrow$                 | translate the view up / down                    |
| left-click + drag                     | rotate (orbit) around the scene                 |
| mouse wheel                           | zoom in / out                                   |
| + / -                                 | zoom in / out from the keyboard                 |
| shift-drag or right-drag              | translate (pan) with the mouse                  |
| Space                                 | start / pause playback                          |
| R, G, T, L, H                         | reset view, toggle grid / trails / labels, help |

The same controls are scriptable from the notebook:

| command                 | effect                                                                                                                                                                                     |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCENE CREATE [port]     | open the window (all entities shown). Default port: one chosen by the OS; give a number (0–65535) to pin it. Prints the URL. A second CREATE is harmless — it just reminds you of the URL. |
| SCENE CLOSE (= DESTROY) | shut the server down; every window disconnects                                                                                                                                             |



| command                    | effect                                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCENE TRANSLATE dx dy [dz] | move the camera's look-at point by (dx, dy, dz) world units (dz defaults to 0)                                                                                                                                                                                                                                                                                                                                  |
| SCENE ROTATE dyaw dpitch   | orbit the camera: yaw (azimuth) and pitch (elevation) in <b>degrees</b> ; pitch is clamped to $\pm 89^\circ$                                                                                                                                                                                                                                                                                                    |
| SCENE ZOOM IN / OUT / f    | zoom by $1.25\times$ , by $1/1.25$ , or by any factor $f > 0$ ( $f > 1$ zooms in)                                                                                                                                                                                                                                                                                                                               |
| SCENE HIDE [n ALL]         | hide object n, or everything (bare HIDE = HIDE ALL)                                                                                                                                                                                                                                                                                                                                                             |
| SCENE SHOW [n ALL]         | undo HIDE                                                                                                                                                                                                                                                                                                                                                                                                       |
| SCENE REFRESH              | copy the notebook's current system into the window (see below) and clear playback history                                                                                                                                                                                                                                                                                                                       |
| SCENE REDRAW               | re-send the complete scene description to every window (forces a full redraw)                                                                                                                                                                                                                                                                                                                                   |
| SCENE START                | begin time-stepped evolution, forward                                                                                                                                                                                                                                                                                                                                                                           |
| SCENE PAUSE                | freeze; START resumes, history is kept                                                                                                                                                                                                                                                                                                                                                                          |
| SCENE STOP                 | halt <b>and clear the recorded history</b>                                                                                                                                                                                                                                                                                                                                                                      |
| SCENE REVERSE              | play <b>backward in time</b> through the recorded history                                                                                                                                                                                                                                                                                                                                                       |
| SCENE RESET                | <b>re-initialize the playback</b> : every mutable value and the time return to their initial values — the state last synced at CREATE/REFRESH, restored bit-identically — history and the step counter clear, and the mode returns to <i>stopped</i> ; START then re-runs the simulation from the beginning. The window's permanent toolbar <b>Reset</b> button does exactly the same (both call one primitive) |
| SCENE SET_TIME_STEP dt     | set the playback time step (must be positive and finite)                                                                                                                                                                                                                                                                                                                                                        |
| SCENE STATUS               | a four-line report: URL + connected windows; mode/t/dt/steps/history; entities + hidden list; camera print (and clear) the asynchronous messages the window has sent: errors, connect/disconnect notices, toolbar actions, data requests                                                                                                                                                                        |
| SCENE EVENTS               |                                                                                                                                                                                                                                                                                                                                                                                                                 |

**The playback copy.** The window animates its *own synchronized copy* of your system, evolved by a background thread — your notebook state does **not** move while the animation runs, so `GET obj0.position` still answers instantly and exactly. The copy is taken at **CREATE** and again at every **REFRESH**. All forward stepping goes through the same SUNDIALS integrators as **STEP/RUN** — there is no separate physics engine in the window.

**Playback is a four-state machine** (*stopped*  $\rightarrow$  *running*  $\rightleftharpoons$  *paused*, plus *reversing*): **STOP** clears history and a later **START** begins fresh; **PAUSE** keeps history so both **START** (forward) and **REVERSE** (backward) can continue from the freeze point. **RESET** is stronger than **STOP**: besides clearing history it puts every value back to its initial state, so the next **START** replays the simulation from the very beginning rather than continuing from wherever playback halted. The reply confirms it:

```
In[6]:= scene reset
Out[6]= scene playback reset to its initial state (t = 0, 1 entity); Start runs the
simulation again from the beginning
```

**How REVERSE works.** While running forward, the playback thread records a snapshot of

the whole system before every step (a ring buffer, at most 20 000 frames). **REVERSE** replays those snapshots newest-first, which is an *exact* rewind — bit-for-bit the states you already visited. When the buffer runs out it pauses and sends an event (**reverse: reached the beginning of recorded history -- paused**). Reversing with no history at all is refused with an error.

**Asynchronous events.** The window can talk back at any time — it reports JavaScript errors, connections, disconnections, and every toolbar action. These messages queue up inside posim; **SCENE EVENTS** drains the queue (up to 1000 messages are kept). In JupyterLab they also appear *by themselves*, without you asking — see §7.

## 5.7 COLLIDE and CONTACTS — rigid-body collisions

|             |                                                     |
|-------------|-----------------------------------------------------|
| COLLIDE     | (report: on/off, collidable pairs, impulses so far) |
| COLLIDE ON  | (default - collisions are detected in EVERY scene)  |
| COLLIDE OFF | (pure point-mass/gravity studies)                   |
| CONTACTS    | (list every contact of the last STEP/RUN)           |

When collisions are on (the default) and the system contains at least one collidable pair (spheres, cuboids, or a point against either — two points cannot collide), **STEP** and **RUN** detect impacts **during** the time step by **SUNDIALS event rootfinding**: the integrator itself lands on the instant where a pair’s signed separation crosses zero, interpolated to solver precision. At that instant an impulse acts along the **contact normal** — the line of the action–reaction force pair — and integration continues. Nothing tunnels, and the reply reports what happened.

Every contact of the last **STEP**/**RUN** is recorded and exposed to the rest of the simulation through read-only **contactK** paths:

| field                                 | type   | meaning                                                                                                                                   |
|---------------------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>contactK.i</b> , <b>contactK.j</b> | number | the colliding pair (indices, $i < j$ )                                                                                                    |
| <b>contactK.t</b>                     | number | the event time (the root the solver landed on)                                                                                            |
| <b>contactK.point</b>                 | vec3   | world-space contact point                                                                                                                 |
| <b>contactK.normal</b>                | vec3   | <b>unit normal, from obji toward objj</b> — the action–reaction line; <b>objj</b> receives $+J\hat{n}$ , <b>obji</b> receives $-J\hat{n}$ |
| <b>contactK.depth</b>                 | number | penetration depth at the event ( $\approx 0$ )                                                                                            |
| <b>contactK.rel_vel_n</b>             | number | approach speed along the normal (negative = approaching)                                                                                  |
| <b>contactK.impulse</b>               | number | scalar impulse magnitude $J$                                                                                                              |

Response physics (full derivation in [collision\\_detection.pdf](#)): per-object **restitution**  $e \in [0, 1]$  (1 = elastic default, 0 = perfectly plastic), pair-combined as  $\min(e_i, e_j)$ ; impulse  $J = -(1 + e)(\mathbf{v}_{\text{rel}} \cdot \hat{n}) / (\hat{n}^\top K \hat{n})$  with the effective-mass matrix  $K$  including the angular terms, applied through the canonical momentum/angular-momentum setters; static bodies (**inverse\_mass** = 0) are immovable walls; approach speeds below **system.restitution\_threshold** settle without bounce; leftover overlap beyond **system.contact\_slop** is projected out. **SCENE** windows draw each contact normal as an arrow at the contact point (**Contacts** toolbar button or the **C** key).

## 5.8 BOX — the rigid bounding box

|            |                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------|
| BOX <size> | (create: six static walls enclosing an inner<br>size x size x size cube centered on the origin) |
| BOX OFF    | (remove the box - its six walls are deleted)                                                    |

|     |                       |
|-----|-----------------------|
| BOX | (bare: report status) |
|-----|-----------------------|

BOX <size> builds a closed rigid room out of **six ordinary cuboid objects** — wall slabs behind the planes  $x = \pm \text{size}/2$ ,  $y = \pm \text{size}/2$ ,  $z = \pm \text{size}/2$  (half-thickness  $\text{size}/4$ , centers at  $\pm 3 \cdot \text{size}/4$ , cross-sections wide enough to cover the corners) — and prints their handles:

|                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>In[2]:= box 4 Out[2]= box: inner size 4 x 4 x 4 -- six static walls obj0, obj1, obj2, obj3, obj4,         obj5 with inverse_mass = 0 (infinitely massive); objects collide elastically off         the inside faces</pre> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

**Infinite mass is exact, not approximate.** The equations of motion never use the mass itself — only the *inverse* mass: velocity is  $v = p m^{-1}$ , and the collision impulse divides by  $\hat{n}^\top K \hat{n} = m_i^{-1} + m_j^{-1} + (\text{angular terms})$  (§5.7). Each wall is created with `inverse_mass = 0` and a zero inverse inertia tensor, so it contributes exactly 0 to every impulse denominator and receives no state writes: bodies bounce off the inside faces elastically while the walls stay **bit-identically at rest**. One measurable consequence: system **momentum is not conserved** inside a box — the infinitely massive walls absorb it without moving (Example 13). LIST tags every wall [`wall: static, inverse_mass=0`] (and shows `mass=0` — the canonical stored quantity for a static body is the inverse).

Bookkeeping:

- GET `system.box` reads the inner side length; 0 means no box. The path is read-only — the box is created and removed by the command.
- A second BOX <size> **replaces** the existing box: the old walls are removed first and the reply notes (replacing the previous box).
- The walls are ordinary objects with ordinary indices: DEL on a wall deletes it like any other object **and dissolves the box** — `system.box` drops to 0 (five walls no longer enclose anything) — but the surviving slabs **stay tracked**: LIST keeps their [`wall: static, inverse_mass=0`] tag, and bare BOX reports `box: dissolved (a wall was deleted; 5 tracked slab(s) remain - BOX <size> replaces them, BOX OFF removes them)`. The next BOX <size> removes the survivors *before* building the new box (the reply notes (replacing the previous box)) and BOX OFF simply deletes them — either way no orphan slab leaks. Wall indices are tracked through every DEL renumbering, so deleting a non-wall object never confuses the box.
- BOX OFF replies `box removed (6 wall(s) deleted, indices renumbered)` — or `box: none` if there was nothing to remove; bare BOX reports the size and wall handles, or `box: none` (BOX <size> creates one).
- A SCENE window draws the box as a **dashed interior wireframe**; the wall slabs themselves are not drawn as bodies. Creating a box while a window is open appends a reminder to the reply: (`scene window open: SCENE REFRESH shows the box`). RESET re-syncs an open window to the now-empty system **and clears its box wireframe and wall flags** — no stale overlay survives the wipe.

## 5.9 User definitions: DEF, LET, FUNCS, SHOW

|                                           |                                     |
|-------------------------------------------|-------------------------------------|
| DEF name(param [= default], ...) { body } | (define a function)                 |
| name(arg, ...)                            | (call it)                           |
| LET name = <expr>                         | (session variable)                  |
| FUNCS                                     | (list signatures; alias: FUNCTIONS) |
| SHOW name                                 | (print a function's source)         |

**DEF is a line form, not a grammar production.** A line that starts with DEF is recognized *before* the ordinary grammar (§3): the notebook captures everything up to the closing } — interactively it keeps prompting ...:= for continuation lines until the brace closes; scripts and JupyterLab cells just keep reading — and installs the function. The **body** is a sequence of ordinary commands, separated by newlines or ;, in which the parameters act as variables. **Every body line is syntax-checked at definition time** (a typo fails the DEF immediately, naming the body line), and **defaults are ordinary expressions evaluated once, at definition** (LET variables are visible in them).

**Calling** uses the ordinary call syntax `name(arg, ...)` — the same atom as `sqrt(2)`; a name that is not a builtin is looked up among your functions. Missing trailing arguments take their defaults (an argument with no default must be supplied). Each call pushes a **call frame** binding the parameters (depth cap: 32 — the language has no conditionals, so recursion could never terminate anyway) and runs the body lines through the same compile-and-execute pipeline as typed input. The call **returns the last body line's value** — if that line is a SET, the call prints no Out[n], exactly like SET itself. A failing body line aborts the call and the error names the function *and* the line; lines that already ran keep their effects (each command's own guarantees still hold per line — e.g. a failing NEW inside a body is still transactional), so a partly-run call leaves no half-built object, only completed commands.

**Editing is SHOW + re-DEF.** SHOW name prints the definition verbatim, exactly as you typed it; redefining the same name replaces the old version and the reply appends (redefined). FUNCS lists every signature with its defaults.

**LET name = expr** binds a session variable (reply: name set). It is visible in bare expressions, in function bodies, in DEF defaults — and, holding a string, it can *name* things: NEW ... AS and named paths resolve a bare identifier through the parameter/LET bindings first (§5.1). Note GET takes only a *path* (§3), so a variable is read back as a bare expression: g0, not GET g0.

A complete session (genuine output):

```
In[1]:= let g0 = 9.81
Out[1]= g0 set
In[2]:= def drop(name, m = 1, h = 10) {
  new sphere as name { mass = m, position = [0, h, 0] }
  set system.gravity = [0, -g0, 0]
}
Out[2]= function drop(3 parameter(s)) defined -- 2 body line(s)
In[3]:= drop("ball")
In[4]:= get ball.position.y
Out[4]= 10
In[5]:= drop("pebble", 0.1, 2)
In[6]:= get pebble.mass
Out[6]= 0.1
In[7]:= funcs
```

```

Out[7]= drop(name, m = 1, h = 10) -- 2 body line(s); SHOW drop prints it
In[8]:= show drop
Out[8]= def drop(name, m = 1, h = 10) {
    new sphere as name { mass = m, position = [0, h, 0] }
    set system.gravity = [0, -g0, 0]
}
In[9]:= def drop(name, m = 1, h = 100) { new sphere as name { mass = m, position = [0,
    h, 0] }; set system.gravity = [0, -g0, 0] }
Out[9]= function drop(3 parameter(s)) defined -- 2 body line(s) (redefined)
In[10]:= drop("skydiver")
In[11]:= skydiver.y
Out[11]= 100
In[12]:= speed
Err[12]: unknown name 'speed' (define it with LET, pass it as a function parameter, or
    use 'speed.field' for a registered object)

```

*What to notice.* The function's **name** parameter holds a string, and **new sphere as name** inside the body names each created object from it — **ball**, **pebble**, **skydiver** — so one definition mints a family of named objects. **drop("ball")** used both defaults; cell 9 is the edit loop (copy cell 8's **SHOW** output, change **h**, re-DEF — note **;** separating body lines works as well as newlines); calls 3, 5 and 10 print no **Out** because the last body line is a **SET**. Cell 12 is the bare-identifier error: execution-time resolution means the message can list every way to bind a name.

Definition-time and call-time errors are specific: a reserved or builtin name is refused (**DEF: 'norm' is a builtin function and cannot be redefined**), nested **DEF** is refused, an empty body is refused, a missing non-default argument fails as **name(): missing argument 'p' (it has no default)**, too many arguments fail naming the signature, and the depth cap fails as **function call depth limit (32) exceeded**.

## 5.10 QM — one-dimensional quantum mechanics

The **QM** family solves the two problems a 1-D quantum solver is for: **bound states** in an arbitrary potential, and **time evolution** of a wavepacket. A session carries one quantum problem, built up command by command.

| command                                         | meaning                                                   |
|-------------------------------------------------|-----------------------------------------------------------|
| QM                                              | report what is currently set up                           |
| QM GRID <x_min> <x_max> <n>                     | the domain and its $n$ interior points                    |
| QM POTENTIAL ZERO                               | free particle                                             |
| QM POTENTIAL BARRIER <v0>, <x1>, <x2>           | $v0$ on $[x_1, x_2]$ , zero elsewhere                     |
| QM POTENTIAL WELL <depth>, <x1>, <x2>           | $-\text{depth}$ on $[x_1, x_2]$                           |
| QM POTENTIAL <function>                         | sample a DEFINED $V(x)$ onto the grid                     |
| QM MASS <m> / QM HBAR <h>                       | both default to 1                                         |
| QM METHOD CAYLEY                                | Crank–Nicolson on the Dirichlet grid — the default        |
| QM METHOD NASH [LIE STRANG]                     | the Bessel-stencil split-operator scheme, <b>periodic</b> |
| QM STATES <k>                                   | the $k$ lowest bound-state energies                       |
| QM STATE <n>                                    | load bound state $n$ as $\psi$                            |
| QM PACKET <x0> <sigma> <k0>                     | a normalised Gaussian wavepacket                          |
| QM STEP <dt> / QM RUN <t> [STEPS <n>]           | propagate with the current QM METHOD                      |
| QM TRANSMISSION <e>                             | $T(E)$ and $R(E)$ at one energy, by transfer matrix       |
| QM SCAN <e1> <e2> <n>                           | scan $T(E)$ , report resonances, push the list            |
| QM NORM / QM ENERGY / QM POSITION / QM MOMENTUM | observables                                               |
| QM PROB <a> <b>                                 | probability of being found in $[a, b]$                    |
| QM DENSITY                                      | $ \psi ^2$ as a list                                      |
| QM DRIVE <shape> <modulation>                   | time-dependent $V(x, t) += f(t)g(x)$                      |
| QM DRIVE OFF                                    | back to a static potential                                |
| QM ABSORB <width> <strength> [<power>]          | absorbing edges; <b>power</b> defaults to 2               |
| QM ABSORB OFF                                   | back to reflecting walls                                  |
| QM ANIMATE "<file>" <t> [FRAMES <n>]            | write a self-contained HTML animation                     |
| QM RESET                                        | forget the quantum problem                                |

Five things are worth knowing before you start, because each will otherwise cost you an afternoon.

**QM METHOD changes the boundary condition, not just the algorithm.** CAYLEY is Crank–Nicolson on the Dirichlet grid: the walls reflect. NASH is the Bessel-stencil split-operator scheme ported from the original C++, and it is **periodic** — a packet leaving the right edge re-enters at the left. The grid points and the potential samples are identical either way, so switching moves nothing; only the two ends change meaning. The status line always states which is in force.

Because bound states are computed with Dirichlet walls, QM STATES and QM STATE are **refused** while the method is NASH rather than quietly handing you eigenstates of a different problem. QM ABSORB and QM DRIVE are likewise refused under NASH: the propagator takes a real, static potential, and an absorber is a complex one. Each refusal names the way out.

NASH alone is first order in  $dt$  — that is what the original does, so it is the default. NASH STRANG is second order for essentially the same cost, and is the better choice unless you are reproducing SolveIt output.

**The walls are infinite and they reflect** — under CAYLEY, which is the default. QM GRID pins  $\psi$  to zero just outside the domain, which is exactly right for bound states and a trap for scattering: a packet that reaches a wall bounces back and corrupts your transmission number. QM PACKET

and QM RUN therefore *watch for it* and print a warning when probability accumulates within 5% of either wall. Take the warning seriously — widen the domain.

**The potential is sampled when the command runs.** QM POTENTIAL *v* evaluates  $v(x)$  at each grid point once and stores the numbers. Editing *v* afterwards changes nothing until you re-issue the command. The status line says so.

**Separate negative arguments with commas.** QM POTENTIAL WELL 5 -2 2 reads 5 - 2 as subtraction and then finds only two arguments where three were wanted. Write 5, -2, 2 — or 5 (-2) 2. Space separation is fine when everything is positive, which is the usual case.

**Why BARRIER and WELL are built in.** This language has no comparison operators, so a piecewise potential cannot be written as a DEF at all — and a square barrier is *the* canonical 1-D problem. They are a deliberate workaround for a language limitation, not a preference for built-ins. Any smooth potential should be a DEF.

## Time-dependent potentials

QM DRIVE makes the potential depend on time, as  $V(x, t) = V_0(x) + f(t)g(x)$ , built from two DEFINED functions.

**Why it is factorised rather than a general  $V(x, t)$ .** A general form would need re-sampling at every grid point on every step — for a user-supplied function, thousands of VM calls per step, costing more than the solve they feed. Factorising costs one evaluation per step and covers the physically important cases exactly: a dipole drive  $f(t)x$ , a shaken trap, a pulse envelope, an adiabatic ramp. A drive whose spatial *profile* changes shape with time is not expressible this way, and that is a real limit rather than an oversight.

The modulation is sampled at the **midpoint** of each step, keeping the scheme second order; sampling at the start would quietly halve it.

Two consequences:

- **Energy is no longer conserved.** A driven system exchanges energy with whatever drives it — physics, not error. QM RUN says so, and the  $\langle E \rangle$  it reports is that of the *static* potential.
- **Propagation is still unitary.**  $H(t)$  is Hermitian at every instant, so each step remains an exact Cayley transform.

For a quadratic potential with a linear drive Ehrenfest's theorem is **exact**:  $\langle x \rangle$  obeys the classical equation of motion with no approximation. Driving the oscillator ground state with  $f(t) = F_0 \cos \omega t$  and  $g(x) = x$  gives

$$x(t) = -\frac{F_0}{1 - \omega^2} [\cos \omega t - \cos t],$$

and the simulation reproduces it — at  $F_0 = 0.3$ ,  $\omega = 0.7$ ,  $t = 5$  the notebook gives  $\langle x \rangle = 0.716918$  against an analytic 0.717840, with the norm at 1.000000000000146.

## Absorbing edges

The reflecting walls are the main practical limit on a scattering run. QM ABSORB replaces them with a **complex absorbing potential**: the Hamiltonian becomes  $H - iW(x)$  with  $W \geq 0$  ramping up smoothly over **width** at each edge, so probability arriving there is *drained* rather than bounced.

Three consequences, all of them reported rather than left to be discovered:

- **Propagation is no longer unitary.** The norm decays. That is the absorber working, so the drift figure printed by QM RUN stops being a health check while it is on.
- **QM STATES is refused.**  $H - iW$  is not Hermitian, and a symmetric eigensolver fed a non-Hermitian matrix returns confident nonsense.
- **The tuning is real.** Too weak and the packet reaches the wall and reflects off *that*; too strong and it reflects off the absorber's own leading edge, because a steep change in the potential is a mirror whether it is real or imaginary. The useful window spans more than an order of magnitude in strength. `cargo run -p quantum --release --example absorber_tuning` measures the whole surface; for  $k_0 \approx 3$  the optimum is near width 18, strength 3, giving about  $10^{-9}$  reflection against a plain wall's 1.0.

A worked comparison: the barrier problem gives  $T = 0.327563$  on a 200-wide reflecting domain, and  $T = 0.327690$  on a **90-wide** domain with QM ABSORB 15 3 — agreement to 4 parts in 10 000 for less than half the grid.

Bound states are found by diagonalising a dense  $n \times n$  matrix with the cyclic Jacobi method, which is  $O(n^3)$ : comfortable to a few hundred points, slow past about a thousand. Propagation is  $O(n)$  per step and has no such limit, so a scattering run can afford a much finer grid than a bound-state calculation.

## 5.11 QM2 — two dimensions, by ADI

A *separate* family from QM, not a mode on it: a 2-D problem differs in almost every argument list, and a hidden mode that silently reinterprets your commands is worse than a second word.

| command                                    | meaning                                     |
|--------------------------------------------|---------------------------------------------|
| QM2                                        | report the current 2-D setup                |
| QM2 GRID <x0> <x1> <nx>, <y0> <y1> <ny>    | domain and resolution per axis              |
| QM2 POTENTIAL ZERO   <function>            | a DEFINED $V(x, y)$                         |
| QM2 PACKET <x0> <y0>, <sx> <sy>, <kx> <ky> | 2-D Gaussian packet                         |
| QM2 STATES <k>                             | the $k$ lowest bound-state energies         |
| QM2 STATE <n>                              | load bound state $n$ as $\psi$              |
| QM2 STEP <dt> / QM2 RUN <t> [STEPS <n>]    | ADI propagation                             |
| QM2 NORM   QM2 ENERGY   QM2 CENTROID       | observables (CENTROID also spells POSITION) |
| QM2 PROB <xa> <xb>, <ya> <yb>              | probability in a rectangle                  |
| QM2 DRIVE <shape> <modulation>   OFF       | time-dependent $V(x, y, t)$                 |
| QM2 ABSORB <w> <s> [<p>]   OFF             | absorbing edges, all four walls             |
| QM2 ANIMATE "<file>" <t> [FRAMES <n>]      | heat-map animation                          |
| QM2 RESET                                  | forget the 2-D problem                      |

### Why ADI, and why not the textbook ADI

In 1-D the Crank–Nicolson operator is tridiagonal, so one solve per step gives an exactly unitary propagator. **In 2-D it is not:** the five-point Laplacian couples each point to neighbours in both



directions and the matrix is block-tridiagonal with bandwidth  $n_x$ , costing  $O(n_x^3 n_y)$  per step to solve directly.

ADI splits the step by direction, so each half-step is a *set of independent tridiagonal solves* —  $n_y$  along  $x$ , then  $n_x$  along  $y$  — and the cost falls to  $O(n_x n_y)$ .

The textbook scheme is **Peaceman–Rachford**, implicit in  $x$  against an explicit  $y$  and then swapped. It is second-order and unconditionally stable, but **not unitary**: the two half-steps use different operators, so the norm drifts at  $O(dt^2)$  per step.

This implementation applies a **Cayley transform in each direction separately**, composed by Strang splitting:

$$\psi(t + dt) = U_x(dt/2) U_y(dt) U_x(dt/2) \psi(t).$$

Each  $A_d = T_d + V/2$  is Hermitian, so each  $U_d$  is exactly unitary and so is their product. **The norm is conserved to machine precision for any  $dt$** , while the splitting error stays  $O(dt^2)$  in the *dynamics*. Norm conservation therefore remains a sharp check on the linear algebra, independent of the accuracy question.

One subtlety: with  $A_x = T_x + V/2$  and  $A_y = T_y + V/2$  the commutator  $[A_x, A_y]$  is non-zero *even for a separable potential*, so a product eigenstate is not perfectly stationary. The drift is second order in  $dt$  and the tests assert that scaling rather than a magnitude.

### Bound states, and why they need a different eigensolver

On an  $n_x \times n_y$  grid the Hamiltonian is  $(n_x n_y)^2$ : a modest  $200 \times 200$  grid gives a  $40\,000 \times 40\,000$  matrix, about 12 GB dense before any arithmetic. The Jacobi solver used in 1-D does not apply.

QM2 STATES uses **Lanczos**, which never forms the matrix — it needs only  $Hv$ , the five-point stencil at  $O(n_x n_y)$ . It builds a Krylov basis, projects  $H$  onto it as a small tridiagonal, and those eigenvalues converge fastest to the *extremes* of the spectrum, which is where bound states are.

Two separate things must go right for degeneracies, and they are easy to confuse:

- **Ghosts.** Lanczos vectors lose orthogonality in floating point as soon as a value converges, and the method rediscovers eigenvalues it already has. Those spurious copies look exactly like degeneracy. Cured by **full reorthogonalisation**.
- **Genuine multiplicity.** Full reorthogonalisation does *not* help. A Krylov space built from one starting vector contains exactly **one** direction from each degenerate eigenspace, so plain Lanczos finds each distinct eigenvalue once however long it runs; no tolerance fixes it. Cured by **deflation**: each converged state is shifted to the top of the spectrum and the solver re-run *from a different starting vector* — reusing the same start would re-derive the direction just removed.

The 2-D isotropic oscillator on a  $70 \times 70$  grid:

```
In[4]:= qm2 states 6
Out[4]= 6 lowest bound state(s) -- Lanczos, 620 iterations:
E[0] = 0.9975639778    residual 3.86e-8
E[1] = 1.9926799032    residual 1.92e-8
E[2] = 1.9926799032    residual 2.89e-8
E[3] = 2.9828813394    residual 3.03e-8
E[4] = 2.9828813394    residual 5.15e-8
```

|                                         |
|-----------------------------------------|
| E[5] = 2.9877958286    residual 5.21e-8 |
|-----------------------------------------|

Against the exact 1, 2, 2, 3, 3, 3. Note what the grid does to the third level:  $E[3]$  and  $E[4]$  agree to **ten digits**, because the square grid keeps the  $x \leftrightarrow y$  symmetry relating the (2, 0) and (0, 2) states — while  $E[5]$ , the (1, 1) state, sits 0.005 away because the continuum *rotational* symmetry that would complete the degeneracy is not a symmetry of the grid. The splitting is discretisation, not solver error.

**Residuals are printed as part of the answer:** an iterative eigensolver has no exact stopping point, and a caller who cannot see how well a state converged cannot know whether to trust it.

Propagation has no comparable limit.

## 5.12 QM3 — three dimensions

A third family, for the same reason QM2 is separate from QM.

| command                                       | meaning                                     |
|-----------------------------------------------|---------------------------------------------|
| QM3 GRID <x0> <x1> <nx>, ..., <z0> <z1> <nz>  | domain and resolution per axis              |
| QM3 POTENTIAL ZERO   <function>               | a DEFINED $V(x, y, z)$                      |
| QM3 PACKET <x0> <y0> <z0>, ...                | 3-D Gaussian packet                         |
| QM3 STATES <k>   QM3 STATE <n>                | bound states                                |
| QM3 STEP <dt>   QM3 RUN <t> [STEPS <n>]       | ADI propagation                             |
| QM3 NORM   QM3 ENERGY   QM3 CENTROID          | observables (CENTROID also spells POSITION) |
| QM3 PROB <xa> <xb>, ..., <za> <zb>            | probability in a box                        |
| QM3 DRIVE <shape> <modulation>   OFF          | time-dependent $V(x, y, z, t)$              |
| QM3 ABSORB <w> <s> [<p>]   OFF                | absorbing faces, all six sides              |
| QM3 ANIMATE "<file>" <t> [FRAMES <n>]         | three marginal densities, animated          |
| QM3 ISO "<file>" <t> [FRAMES <n>] [LEVEL <f>] | rotatable isosurface                        |
| QM3 RESET                                     | forget the 3-D problem                      |

The scheme is the 2-D one with a third direction, Strang-composed so every factor stays an exact Cayley transform:

$$\psi(t + dt) = U_x(dt/2) U_y(dt/2) U_z(dt) U_y(dt/2) U_x(dt/2) \psi(t),$$

with  $A_d = T_d + V/3$ . Five directional sweeps per step instead of three, each a set of independent tridiagonal solves, so a step is still  $O(n_x n_y n_z)$  and **the norm is conserved to machine precision for any  $dt$ .**

### What changes in three dimensions is memory

A  $100^3$  grid is a million points. One complex wavefunction is 16 MB, which is fine — but the 2-D propagator precomputes full-size band arrays, three per direction, and doing that here would cost about 140 MB before any work began. QM3 builds its bands **per line, on the fly**, so the only extra storage is  $O(\max(n_x, n_y, n_z))$ .

## The eigensolver has a ceiling, and says so

Propagation is comfortable past  $64^3$ . QM3 STATES is not: Lanczos reorthogonalises fully and stores its whole Krylov basis, costing  $O(m^2n)$  in time and  $O(mn)$  in memory. The practical ceiling is about  $40^3$ , and beyond it the command **refuses up front** rather than exhausting memory. Propagation on the same grid still works.

## Looking at a volume

A volume cannot be drawn on a flat canvas without an isosurface mesh or a ray-caster, either of which would mean shipping a WebGL pipeline inside a file that must work from `file://`. QM3 ANIMATE shows the three **marginal densities** instead:

$$P(x, y) = \int |\psi|^2 dz, \quad P(x, z) = \int |\psi|^2 dy, \quad P(y, z) = \int |\psi|^2 dx.$$

Each is a genuine observable — the probability of finding the particle at those two coordinates whatever the third — not a rendering convention, and each integrates to the total norm, which the page prints under every panel.

What they cannot show is correlation between axes: a state concentrated on a diagonal shell and one spread over a box can share all three marginals. That is a real limit of the representation, and the reason a true isosurface view is listed as unfinished rather than unnecessary.

Verified in a browser on a driven  $34^3$  run: all three marginals integrate to 1.000000 in every sampled frame, the  $P(x, y)$  peak travels along  $x$  while its  $y$  coordinate does not move, and  $P(y, z)$  does not move at all — exactly right for a drive along  $x$ .

## Isosurfaces

Marginals cannot show correlation between axes; an isosurface can. QM3 ISO extracts the surface where  $|\psi|^2$  equals a chosen fraction of its peak.

**Marching tetrahedra, not marching cubes.** Marching cubes needs a 256-entry table mapping corner-sign patterns to triangle lists — and that table *is* the algorithm, so one wrong entry gives a surface with a hole that looks fine from most angles. Marching tetrahedra needs no table: six tetrahedra per cube,  $2^4 = 16$  sign patterns, three cases up to symmetry. The classic ambiguity where two cubes triangulate a shared face incompatibly cannot arise, because a tetrahedron’s faces are triangles.

The extractor is a library function with quantitative tests: for a sphere and an ellipsoid it checks enclosed volume by the divergence theorem, surface area, convergence under refinement, and that the mesh is **watertight and consistently oriented** — every directed edge exactly once, its reverse exactly once.

Two defects that testing found rather than inspection: samples lying *exactly* on the isosurface produced coincident vertices and pinholes (fixed by nudging the level, not the samples); and orienting a split quad’s two triangles independently let one flip and open the shared diagonal (they now share one winding decision).

Rendering is a **software rasteriser on a 2-D canvas**, not WebGL: WebGL can fail silently without a GPU and then the page shows nothing, whereas a painter’s-algorithm rasteriser works everywhere and can be verified by reading pixels back.

## The 3-D oscillator

```
In[4]:= qm3 states 4
Out[4]= 4 lowest bound state(s) -- Lanczos, 245 iterations:
  E[0] = 1.4738115042    residual 5.71e-8
  E[1] = 2.4382167269    residual 2.51e-8
  E[2] = 2.4382167269    residual 5.03e-8
  E[3] = 2.4382167269    residual 5.69e-8
```

Against the exact  $E = n_x + n_y + n_z + 3/2$ :  $3/2$ , then  $5/2$  **three times**. The three excited values agree to ten digits — that degeneracy is what deflation exists to resolve — while both levels sit about 0.03 below the continuum, which is second-order discretisation error at  $h = 0.52$ .

## 6 The notebook (cells, editing, magics)

### 6.1 Cells

Start `posim` with no arguments. You get numbered prompts, exactly like Jupyter or Mathematica:

```
In[1]:= new sphere { mass = 2, radius = 0.5 }
Out[1]= obj0
In[2]:= get obj0.inertia_tensor
Out[2]= [[0.2, 0, 0], [0, 0.2, 0], [0, 0, 0.2]]
```

Pressing **Enter** executes the cell — in a plain terminal, Enter *is* shift-enter. Commands with no interesting result (like `SET`) print no `Out[n]`. Errors print `Err[n]:` and the session continues.

### 6.2 State is cumulative

Like Jupyter, the notebook has one live simulator state that every cell mutates in order. Re-running an old `NEW` cell creates a *new* object; it does not replace the old one. To rebuild cleanly, `%reset` and replay.

### 6.3 Magics

| magic                             | effect                                                           |
|-----------------------------------|------------------------------------------------------------------|
| <code>%history</code>             | lists every cell with input and output; failed cells marked !    |
| <code>%edit n &lt;text&gt;</code> | replaces cell <i>n</i> 's input and executes it as the next cell |
| <code>%rerun n</code>             | executes cell <i>n</i> 's input again as a new cell              |
| <code>%save &lt;file&gt;</code>   | writes all successful non-magic inputs to a replayable script    |
| <code>%load &lt;file&gt;</code>   | replays a script file cell by cell                               |
| <code>%reset</code>               | clears the simulator (history kept)                              |
| <code>%quit, %exit</code>         | leave                                                            |

A plain terminal has no cursor-addressable cells (`posim` uses only the Rust standard library — no raw terminal mode), so backward/forward movement is by these magics. **For true click-and-edit cells with shift-enter, use the JupyterLab front end (§7).**

## 6.4 Scripts

`posim -script file` executes a file of command lines (blank lines and `#` comments skipped), echoing `In[n]/Out[n]`, and exits nonzero if any cell failed — good for CI.

`posim -notebook file` executes the same file format but then **stays in the interactive notebook** instead of exiting: the loaded cells keep their `In[n]` numbers, the next command continues the numbering, and a scene window the file opened (`SCENE CREATE`) stays alive — the launcher for the *dynamic notebooks* in `dynamic_notebooks/`, files that build a simulation and open the GUI ready for Start. A failing cell aborts the load with a nonzero exit instead of entering the notebook.

## 7 JupyterLab in one paragraph

The `jupyter/` directory ships a small *wrapper kernel*: JupyterLab starts a Python shim, the shim starts `posim -machine` (a JSON request/reply protocol over stdin/stdout), and each notebook cell you shift-enter is forwarded line-by-line as `{"op":"exec","code":...}`. Setup and tests: `jupyter/README.md`. Everything in §§2–5 applies verbatim inside JupyterLab cells; multi-line cells run one line at a time, stopping at the first error.

## 8 The stack machine (what runs your line)

The parser emits *postfix instructions*; e.g. `set obj0.mass = 2 + 3 * 4` compiles to

|                 |                                |
|-----------------|--------------------------------|
| Push 2          |                                |
| Push 3          |                                |
| Push 4          |                                |
| Mul             | <- pops 3,4 pushes 12          |
| Add             | <- pops 2,12 pushes 14         |
| Store obj0.mass | <- pops 14, calls set_mass(14) |

The machine’s whole memory is a stack of typed values. Instructions: `Push`, `Load`, `Store`, `LoadIdent` (a bare identifier — parameter or LET variable, resolved at execution, §5.9), `StoreGlobal` (LET), `Add` `Sub` `Mul` `Div` `Neg`, `PackList` (literals), `Call` (builtin or user function — a user call runs each body line through this same compile-and-execute pipeline inside a call frame), `NewObject/InitField/FinishM` (which also registers the AS name), `Delete`, `ListObjects`, `ListFns/ShowFn` (FUNCS/SHOW), `Step`, `Run`, `SetMethod`, `Energy`, `CenterOfMass`, `TotalMomentum`, `TotalAngularMomentum`, `Laplace`, `Reset`, `Help`. `Load/Store` go *only* through the `physical_object` get/set API — the machine cannot corrupt simulator state, and every coupled invariant (`mass↔inverse`, `inertia↔inverse`, unit quaternions) is enforced on every write. When the program ends, the top of the stack becomes `Out[n]`.

## 9 Eighteen worked examples

All transcripts are genuine program output.

### 9.1 Example 1 — an eccentric Kepler orbit and the Runge–Lenz compass

The Laplace–Runge–Lenz vector **A** points along a Kepler orbit’s major axis and is conserved *only* for a perfect  $1/r^2$  force — the most delicate conservation test available. We build a “sun” so heavy the reduced problem is exact and integrate two orbits with a 4th-order *symplectic* method.

```

In[1]:= new point { mass = 1e9, position = [0, 0, 0] }
Out[1]= obj0
In[2]:= new point { mass = 1, position = [0.4, 0, 0], velocity = [0, 2, 0] }
Out[2]= obj1
In[3]:= set system.g_constant = 1e-9
In[4]:= set system.softening = 0
In[5]:= laplace 1
Out[5]= [0.5999999974000001, 0, 0]
In[6]:= method sprk mclachlan_4_4 0.001
Out[6]= method = ARKODE SPRK ARKODE_SPRK_MCLACHLAN_4_4, fixed dt = 0.001
In[7]:= run 12.6 steps 2
Out[7]= t = 12.6 (12600 solver steps, 2 snapshots, |dE/E| = 9.237e-14)
In[8]:= laplace 1
Out[8]= [0.5999999974140128, -0.00000000034051644837163053, 0]
In[9]:= get obj1.position
Out[9]= [0.3964802853115723, 0.06706198328283366, 0]

```

*What to notice.* `g_constant = 1e-9` makes  $GM_{\text{total}} = 1$  so the orbit has period  $2\pi$ . LAPLACE 1 divided by `mk` is the *eccentricity vector*:  $e = 0.6$  is read directly off `Out[5]`. After  $\sim 2$  orbits the energy is conserved to  $9 \times 10^{-14}$  and **A** has rotated by only  $3 \times 10^{-10}$  — no spurious perihelion precession. Softening was zeroed because any  $\varepsilon \neq 0$  slightly breaks the  $1/r^2$  law and *would* precess the axis.

## 9.2 Example 2 — a thrown ball vs. the textbook formula

A projectile has the closed form  $x(t) = v_x t$ ,  $y(t) = y_0 + v_y t - \frac{1}{2}gt^2$ . We fly a baseball for 6.12 s and subtract the formula *inside the notebook* using bare expressions.

```

In[1]:= new point { mass = 0.145, position = [0, 1, 0], velocity = [30, 30, 0] }
Out[1]= obj0
In[2]:= set system.gravity = [0, -9.81, 0]
In[3]:= run 6.12 steps 3
Out[3]= t = 6.12 (13 solver steps, 3 snapshots, |dE/E| = 4.740e-15)
In[4]:= get obj0.position.x
Out[4]= 183.59999999999999
In[5]:= 30 * 6.12
Out[5]= 183.6
In[6]:= obj0.position.x - 30 * 6.12
Out[6]= -0.00000000000008526512829121202
In[7]:= obj0.position.y - (1 + 30 * 6.12 - 9.81 / 2 * 6.12 * 6.12)
Out[7]= -0.0000000000004263256414560601

```

*What to notice.* Cells 6–7 are *bare expressions* mixing paths and arithmetic (GET itself would refuse the arithmetic). The trajectory matches the parabola to  $\sim 10^{-13}$  — and the adaptive Adams integrator needed only **13 internal steps**, because for a polynomial solution its error estimator lets it take huge strides.

## 9.3 Example 3 — cyclotron motion: expressions as arguments

A charge in a uniform magnetic field circles with period  $T = 2\pi m/(|q|B)$ . We *compute*  $T$  *inside the RUN command*.

```

In[1]:= new sphere { mass = 2, radius = 0.1, charge = -1.5, velocity = [3, 0, 0] }
Out[1]= obj0
In[2]:= set system.b_field = [0, 0, 4]
In[3]:= method bdf
Out[3]= method = CVODE BDF
In[4]:= 2 * pi * 2 / (1.5 * 4)
Out[4]= 2.0943951023931953
In[5]:= run 2 * pi * 2 / (1.5 * 4) steps 8
Out[5]= t = 2.0943951023931953 (318 solver steps, 8 snapshots, |dE/E| = 1.444e-8)
In[6]:= get obj0.position
Out[6]= [0.00000000043106769672882073, 0.000000007220835657713453, 0]
In[7]:= norm(obj0.velocity)
Out[7]= 2.9999999783374913

```

*What to notice.* After one analytic period the particle is back at the origin to  $7 \times 10^{-9}$ , and `norm()` shows the speed unchanged — the magnetic force does no work. The Lorentz force  $q\mathbf{v} \times \mathbf{B}$  is velocity-dependent, so the symplectic path is *not allowed* here; BDF is the right tool for fast gyration.

#### 9.4 Example 4 — the tennis-racket theorem (Dzhanibekov effect)

A rigid body spun about its *intermediate* principal axis is unstable: a tiny wobble grows into a dramatic flip, yet energy and angular momentum stay conserved. Half-extents [0.5, 1, 2] give moments [5, 4.25, 1.25]; spinning about  $y$  (the middle value) triggers it.

```

In[1]:= new cuboid { mass = 3, half_extents = [0.5, 1, 2], angular_velocity = [0.01, 3,
    0.01] }
Out[1]= obj0
In[2]:= get obj0.inertia_tensor
Out[2]= [[5, 0, 0], [0, 4.25, 0], [0, 0, 1.25]]
In[3]:= get obj0.angular_velocity
Out[3]= [0.010000000000000002, 3, 0.010000000000000002]
In[4]:= run 40 steps 4
Out[4]= t = 40 (2361 solver steps, 4 snapshots, |dE/E| = 5.605e-9)
In[5]:= get obj0.angular_velocity
Out[5]= [1.5428438559953521, 2.9947703802651175, -0.7871804456905063]
In[6]:= run 40 steps 4
Out[6]= t = 80 (2334 solver steps, 4 snapshots, |dE/E| = 5.332e-9)
In[7]:= get obj0.angular_velocity
Out[7]= [0.020666590902422656, 2.9999548562146483, 0.013346831306280834]
In[8]:= get obj0.angular_momentum
Out[8]= [0.05, 12.75, 0.0125]

```

*What to notice.* These two samples are snapshots of a *recurring* motion, not the start and end of one flip. The turn-over is a *body-frame* event: the angular momentum is fixed in space, so the world-frame  $\omega_y$  printed above stays near +3 for the whole run and never reverses at all. Extract  $\omega_{\text{body}} = \bar{q}\omega_{\text{world}}q$  and its  $y$ -component reverses at roughly  $t = 4, 18, 28, 40, 50, 62, 74$  — about every 12–13 units. By  $t = 80$  the body has turned over some *seven* times;  $t = 40$  merely lands mid-reversal and  $t = 80$  between reversals. Throughout, the world-frame angular momentum (Out[8]) is exactly the initial  $I\omega = [0.05, 12.75, 0.0125]$  — the solver integrates the full quaternion

+ angular-momentum state with energy error  $\sim 5 \times 10^{-9}$ . (With these moments a symmetric wobble  $(d, 3, d)$  sits exactly on the separatrix  $G^2 = BT$ , since  $G^2 - BT = d^2[A(A-B) + C(C-B)] = d^2[5(0.75) + 1.25(-3)] = 0$  for every  $d$ ; the repeated reversal is robust, but the exact times shift with the tolerances. `dynamic_notebooks/tumbling_body.posim` shows the body-frame component directly.)

## 9.5 Example 5 — three bodies, then surgery with DEL

```
In[1]:= new point { mass = 1,   position = [1, 0, 0],   velocity = [0, 1, 0] }
Out[1]= obj0
In[2]:= new point { mass = 4,   position = [-1, 0, 0], velocity = [0, -0.25, 0] }
Out[2]= obj1
In[3]:= new point { mass = 256, position = [0, 8, 0],   velocity = [0, 0, 0] }
Out[3]= obj2
In[4]:= set system.g_constant = 0.001
In[5]:= momentum
Out[5]= [0, 0, 0]
In[6]:= com
Out[6]= [-0.011494252873563218, 7.846743295019157, 0]
In[7]:= run 3 steps 3
Out[7]= t = 3 (84 solver steps, 3 snapshots, |dE/E| = 3.141e-10)
In[8]:= list
Out[8]= obj0: point, mass=1, charge=0, pos=[0.9936356911262184, 3.022315903943394, 0]
        obj1: point, mass=4, charge=0, pos=[-0.9972663866755148, -0.7330950497234504, 0]
        obj2: point, mass=256, charge=0, pos=[-0.000017852126656859762, 7.999648688652149, 0]
In[9]:= del 2
Out[9]= deleted obj2; 2 object(s) remain (indices renumbered)
In[10]:= list
Out[10]= obj0: point, mass=1, charge=0, pos=[0.9936356911262184, 3.022315903943394, 0]
        obj1: point, mass=4, charge=0, pos=[-0.9972663866755148, -0.7330950497234504, 0]
In[11]:= momentum
Out[11]= [0.0021430470372574067, 0.061528401914275554, 0]
```

*What to notice.* Cells 1–3 chose velocities so the initial total momentum is exactly zero ( $1 \cdot 1 + 4 \cdot (-0.25) = 0$ ), and it stays zero through the run. Deleting the heavy body removes its momentum: the remainder (Out[11]) is what the surviving pair carries — conservation applies to the system you keep. DEL renumbers later objects.

## 9.6 Example 6 — magnetic torque (and why ENERGY grows)

magnetic\_moment\_tensor  $M$  couples the world  $B$  field to torque  $\tau = (RMR^T)B$ . Unlike everything else, this coupling is *not derived from a potential*, so total energy is *expected* to change.

```
In[1]:= new sphere { mass = 1, radius = 0.5, magnetic_moment_tensor = [[0.2, 0, 0], [0,
        0.2, 0], [0, 0, 0.2]] }
Out[1]= obj0
In[2]:= set system.b_field = [0, 0.5, 0]
In[3]:= get obj0.angular_momentum
Out[3]= [0, 0, 0]
In[4]:= energy
Out[4]= 0
In[5]:= run 4 steps 4
```



```

Out[5]= t = 4 (214 solver steps, 4 snapshots, |dE/E| = 8.000e-1)
In[6]:= get obj0.angular_momentum
Out[6]= [0, 0.400000000000000085, 0]
In[7]:= get obj0.angular_velocity
Out[7]= [0, 4.0000000000000009, 0]
In[8]:= energy
Out[8]= 0.80000000000000035

```

*What to notice.* Constant torque  $\tau = MB = 0.2 \times 0.5 = 0.1$  about  $y$  gives  $L(t) = 0.1t$ : after 4 s,  $L = 0.4$  exactly (the ODE is linear, so the solver nails it). Sphere inertia  $= 0.4 \cdot 1 \cdot 0.25 = 0.1$ , so  $\omega = L/I = 4$  and the rotational energy  $\frac{1}{2}\omega L = 0.8$  matches Out[8]. The reported  $|dE/E| = 8e-1$  is not an error — it honestly reports that the torque pumped energy in.

## 9.7 Example 7 — fixing a typo in an old cell with %edit

```

In[1]:= new sphere { mass = 20, radius = 0.5 }
Out[1]= obj0
In[2]:= set obj0.velocity = [1, 0, 0]
In[3]:= get obj0.momentum
Out[3]= [20, 0, 0]
In[4]:= %edit 1 new sphere { mass = 2, radius = 0.5 }
In[4]:= new sphere { mass = 2, radius = 0.5 }
Out[4]= obj1
In[5]:= %history
In[1]:= new sphere { mass = 2, radius = 0.5 }
Out[1]= obj0
In[2]:= set obj0.velocity = [1, 0, 0]
In[3]:= get obj0.momentum
Out[3]= [20, 0, 0]
In[4]:= new sphere { mass = 2, radius = 0.5 }
Out[4]= obj1

```

*What to notice.* `%edit 1 ...` rewrites cell 1's stored input **and re-executes it as the next cell** — like editing a Jupyter cell and shift-entering again. Because state is cumulative, the re-executed **NEW** made a *second* object; the fat `obj0` is still there. For a clean rebuild use `%reset + %rerun` or `%load`; when a field tweak suffices, `set obj0.mass = 2`. `%history` shows the *edited* input but the *original* outputs — an audit trail of what actually ran.

## 9.8 Example 8 — checking the simulator against itself

```

In[1]:= new point { mass = 2, position = [1, 0, 0], velocity = [0, 3, 0] }
Out[1]= obj0
In[2]:= cross(obj0.position, obj0.momentum)
Out[2]= [0, 0, 6]
In[3]:= angmom
Out[3]= [0, 0, 6]
In[4]:= dot(obj0.position, obj0.velocity)
Out[4]= 0
In[5]:= norm(cross(obj0.position, obj0.momentum)) / (norm(obj0.position) * norm(obj0.
momentum))
Out[5]= 1

```

*What to notice.* Cell 2 computes  $L = r \times p$  by hand and cell 3 confirms ANG MOM agrees. Cell 4 proves  $r \perp v$ ; cell 5 computes  $|r \times p|/(|r||p|) = \sin \theta = 1$  — the angle between  $r$  and  $p$  is  $90^\circ$  — inside one nested expression.

## 9.9 Example 9 — hand-built tensors and a quaternion orientation

```
In[1]:= new cuboid { mass = 1, inertia_tensor = [[2, 0, 0], [0, 3, 0], [0, 0, 4]],
orientation = [0.7071067811865476, 0, 0, 0.7071067811865476] }
Out[1]= obj0
In[2]:= get obj0.inverse_inertia_tensor
Out[2]= [[0.5, 0, 0], [0, 0.3333333333333333, 0], [0, 0, 0.25]]
In[3]:= set obj0.angular_velocity = [0, 2, 0]
In[4]:= get obj0.angular_momentum
Out[4]= [0.00000000000000004440892098500628, 4.0000000000000002, 0]
In[5]:= get obj0.orientation
Out[5]= quat[w=0.7071067811865476, x=0, y=0, z=0.7071067811865476]
```

*What to notice.* Because `inertia_tensor` appears in the initializer, the analytic inertia is **not** recomputed — your `diag(2,3,4)` is kept, with the inverse maintained automatically. The quaternion  $[w, x, y, z] = [\sqrt{1/2}, 0, 0, \sqrt{1/2}]$  is a  $90^\circ$  rotation about  $z$ , so setting the *world* angular velocity  $[0, 2, 0]$  exercises the full  $L = (RIR^T)\omega$  transformation. By hand: rotating `diag(2,3,4)` by  $90^\circ$  about  $z$  swaps the  $x$  and  $y$  moments, so  $RIR^T = \text{diag}(3, 2, 4)$ , the world- $y$  moment is 2, and  $L = [0, 4, 0]$  — exactly `Out[4]` (up to  $4 \times 10^{-16}$  of floating-point dust).

## 9.10 Example 10 — reproducibility: %save, %reset, %load

```
In[1]:= new sphere { mass = 2, radius = 0.5, velocity = [1, 0, 0] }
Out[1]= obj0
In[2]:= set system.gravity = [0, -9.81, 0]
In[3]:= step 0.5
Out[3]= t = 0.5 (advanced by 0.5, 12 solver steps)
In[4]:= %save session.posim
saved 3 cell(s) to session.posim
In[4]:= %reset
system reset
In[4]:= list
Out[4]= (no objects)
In[5]:= %load session.posim
In[5]:= new sphere { mass = 2, radius = 0.5, velocity = [1, 0, 0] }
Out[5]= obj0
In[6]:= set system.gravity = [0, -9.81, 0]
In[7]:= step 0.5
Out[7]= t = 0.5 (advanced by 0.5, 12 solver steps)
In[8]:= get obj0.velocity
Out[8]= [1, -4.905000000000001, 0]
```

*What to notice.* `%save` writes only the successful, non-magic inputs — a clean, replayable script. After `%reset`, `%load` replays it and the state is bit-identical ( $v_y = -9.81 \cdot 0.5 = -4.905$ ). The same file runs headlessly via `posim -script`.

Notice too that **magics do not consume cell numbers**: `%save` and `%reset` print their message

and leave the counter alone, which is why the prompt shows `In[4]` three times in a row before `LIST` finally becomes cell 4. Only executed commands become numbered cells — so the `In[n]` numbering always matches what `%history` will replay.

The machine mode speaks the same language over JSON:

```
$ printf '%s\n' '{"op":"exec","code":"new point { mass = 1, velocity = [0, 1, 0] }"}' \
                '{"op":"get","path":"obj0.momentum"}' \
                '{"op":"set","path":"obj0.mass","value":5}' \
                '{"op":"get","path":"obj0.momentum"}' | posim --machine
{"display":"obj0","ok":true,"result":"obj0"}
{"display":"[0, 1, 0]","ok":true,"result":[0.0,1.0,0.0]}
{"display":"","ok":true,"result":null}
{"display":"[0, 1, 0]","ok":true,"result":[0.0,1.0,0.0]}
```

Changing the mass did **not** change the momentum — momentum is the canonical stored state; the *velocity* is what changed. That final subtlety is the union design showing through the wire protocol.

## 9.11 Example 11 — watching a Kepler orbit live, then running it backward

The same two-body setup as Example 1 — but this time we *watch* it. `SCENE CREATE` opens the scene window in the browser; `SCENE START` sets it in motion; `SCENE REVERSE` runs the movie backward.

```
In[1]:= new point { mass = 1e9, position = [0, 0, 0] }
Out[1]= obj0
In[2]:= new point { mass = 1, position = [0.4, 0, 0], velocity = [0, 2, 0] }
Out[2]= obj1
In[3]:= set system.g_constant = 1e-9
In[4]:= set system.softening = 0
In[5]:= scene create 7878
Out[5]= scene window created: http://127.0.0.1:7878/
(opened in your browser; if no window appeared, open that address yourself)
showing 2 entities; SCENE START begins the evolution -- HELP lists all scene commands
In[6]:= scene set_time_step 0.005
Out[6]= scene time step dt = 0.005
In[7]:= scene start
Out[7]= scene playback: running
In[8]:= scene pause
Out[8]= scene playback: paused
In[9]:= scene status
Out[9]= scene: http://127.0.0.1:7878/ (1 window(s) connected)
mode = paused, t = 0.4550000000000003, dt = 0.005, steps = 91, history = 91 frame(s)
entities = 2 (hidden: none)
camera: yaw = -60°, pitch = 55°, dist = 12, target = [0, 0, 0]
In[10]:= scene reverse
Out[10]= scene playback: reversing
In[11]:= scene status
Out[11]= scene: http://127.0.0.1:7878/ (1 window(s) connected)
mode = reversing, t = 0.15500000000000005, dt = 0.005, steps = 91, history = 31 frame(s)
)
entities = 2 (hidden: none)
```

```

camera: yaw = -60°, pitch = 55°, dist = 12, target = [0, 0, 0]
In[12]:= scene events
Out[12]= window connected (1 total)
In[13]:= scene close
Out[13]= scene closed (http://127.0.0.1:7878/)

```

*What to notice.* Between In[7] and In[8] a few real seconds passed while the orbit ran in the window — playback happens on a background thread at  $\sim 30$  frames per second, so **the notebook prompt never blocks**: cell 9’s STATUS shows the *playback copy* has advanced 91 steps to  $t \approx 0.455$  while your notebook state still sits untouched at  $t = 0$  (that is the “playback copy” design: the window animates its own synchronized copy of your system, so GET `system.time` here would still print 0). After REVERSE (cell 10), the status in cell 11 shows time flowing *backward* ( $t \approx 0.155$ ) and the history buffer draining ( $91 \rightarrow 31$  frames): each reversed frame is an exact recorded snapshot, so the rewind retraces the orbit bit-for-bit. Cell 12 drains the asynchronous event queue — the window announced itself when the browser connected. In JupyterLab you would not even need cell 12: events surface on their own as [scene] ... lines (§7).

## 9.12 Example 12 — driving the camera from the notebook

Every gesture the mouse can make in the window has a command twin, so a *script* can compose the exact view you want — useful for repeatable demonstrations and screenshots.

```

In[1]:= new sphere { mass = 3, radius = 0.6, position = [2, 0, 0] }
Out[1]= obj0
In[2]:= new cuboid { mass = 1, half_extents = [0.4, 0.3, 0.2], position = [-2, 0, 1] }
Out[2]= obj1
In[3]:= scene create 7878
Out[3]= scene window created: http://127.0.0.1:7878/
(opened in your browser; if no window appeared, open that address yourself)
showing 2 entities; SCENE START begins the evolution -- HELP lists all scene commands
In[4]:= scene translate 2 0
Out[4]= camera target = [2, 0, 0]
In[5]:= scene rotate 30 -10
Out[5]= camera yaw = -30°, pitch = 45°
In[6]:= scene zoom in
Out[6]= camera distance = 9.6
In[7]:= scene zoom 2
Out[7]= camera distance = 4.8
In[8]:= scene zoom out
Out[8]= camera distance = 5.999999999999999
In[9]:= scene hide 0
Out[9]= 1 object(s) hidden
In[10]:= scene show all
Out[10]= 0 object(s) hidden
In[11]:= scene redraw
Out[11]= scene redraw queued for every window
In[12]:= scene status
Out[12]= scene: http://127.0.0.1:7878/ (0 window(s) connected)
mode = stopped, t = 0, dt = 0.01, steps = 0, history = 0 frame(s)
entities = 2 (hidden: none)
camera: yaw = -30°, pitch = 45°, dist = 5.999999999999999, target = [2, 0, 0]
In[13]:= scene close

```

```
Out[13]= scene closed (http://127.0.0.1:7878/)
```

*What to notice.* The camera is an **orbit camera**: it circles a *look-at point* at a *distance*, aimed by *yaw* (compass direction) and *pitch* (elevation). `TRANSLATE 2 0` moves the look-at point onto the sphere (this is what the arrow keys do in the window); `ROTATE 30 -10` adds  $30^\circ$  of yaw to the default  $-60^\circ$  and  $-10^\circ$  of pitch to the default  $55^\circ$  (this is what left-dragging does); the three zooms multiply the distance by  $1/1.25$ , then  $1/2$ , then  $1.25$  — watch the arithmetic:  $12 \rightarrow 9.6 \rightarrow 4.8 \rightarrow 6$  (this is the mouse wheel). `HIDE 0` blanks the sphere out of every connected window without deleting anything — `SHOW ALL` brings it back. Because this transcript ran headless, cell 12 reports `0 window(s) connected` — the commands work regardless, and any window that connects later receives the composed view. Note also the two-argument form `scene translate 2 0`: the optional `dz` defaulted to 0, and `-10` in cell 5 was one negative argument, not a subtraction (scene arguments are *terms* — the second subtlety of §3).

### 9.13 Example 13 — the box of shapes: every body type in a rigid, infinitely massive box

The finale scene: `BOX 4` builds the rigid room (§5.8) and one of every shape goes inside ( $R = 1$ ,  $M = 1$ ) — a **torus** (mass  $M$ , inner radius 1, outer radius 2), a **point** ( $M$ , the only mover:  $v = (100, 200, 100)$ ), a **sphere** ( $2M$ ,  $r = \frac{1}{2}$ ), an ideal zero-thickness **disk** ( $2M/3$ ,  $r = 1$ ), a **cube** ( $5M/3$ , side 1) and a **cylinder** ( $2M$ ,  $r = \frac{1}{2}$ , height  $3/2$ ). An axis-aligned torus of outer radius 2 would *exactly inscribe* the 4-box (its outer equator touching four walls), so the torus is tilted with its axis along  $(1, 1, 1)/\sqrt{3}$ : its extent per axis is  $1.5\sqrt{2/3} + 0.5 \approx 1.7247 < 2$  — clearance 0.2753. The other positions are random, drawn by a documented LCG ( $x \leftarrow 1664525x + 1013904223 \bmod 2^{32}$ , seed 20260724, sequential rejection at  $\geq 0.05$  separation) — see `scripts/collisions/11_box_of_shapes.posim`, the script this transcript replays.

```
In[1]:= set system.g_constant = 0
In[2]:= box 4
Out[2]= box: inner size 4 x 4 x 4 -- six static walls obj0, obj1, obj2, obj3, obj4,
        obj5 with inverse_mass = 0 (infinitely massive); objects collide elastically off
        the inside faces
In[3]:= new torus { mass = 1, inner_radius = 1, outer_radius = 2, orientation =
        [0.888073833977115, -0.325057583671868, 0.325057583671868, 0] }
Out[3]= obj6
In[4]:= new point { mass = 1, position = [1.406590, -0.995859, 0.569601], velocity =
        [100, 200, 100] }
Out[4]= obj7
In[5]:= new sphere { mass = 2, radius = 1/2, position = [1.424704, 1.367496, -0.493612]
        }
Out[5]= obj8
In[6]:= new disk { mass = 2/3, radius = 1, position = [-0.677386, -1.041493,
        -1.091679], orientation = [0.900447102352677, 0.307567078752479,
        -0.307567078752479, 0] }
Out[6]= obj9
In[7]:= new cuboid { mass = 5/3, half_extents = [0.5, 0.5, 0.5], position = [1.074397,
        0.816223, 1.102099] }
Out[7]= obj10
In[8]:= new cylinder { mass = 2, radius = 1/2, height = 3/2, position = [-1.027024,
        -1.403890, -0.485619], orientation = [0.968912421710645, 0, 0.247403959254523, 0] }
Out[8]= obj11
```

```

In[9]:= list
Out[9]= obj0: cuboid he=[1, 4, 4], mass=0, charge=0, pos=[3, 0, 0] [wall: static,
      inverse_mass=0]
obj1: cuboid he=[1, 4, 4], mass=0, charge=0, pos=[-3, 0, 0] [wall: static, inverse_mass
      =0]
obj2: cuboid he=[4, 1, 4], mass=0, charge=0, pos=[0, 3, 0] [wall: static, inverse_mass
      =0]
obj3: cuboid he=[4, 1, 4], mass=0, charge=0, pos=[0, -3, 0] [wall: static, inverse_mass
      =0]
obj4: cuboid he=[4, 4, 1], mass=0, charge=0, pos=[0, 0, 3] [wall: static, inverse_mass
      =0]
obj5: cuboid he=[4, 4, 1], mass=0, charge=0, pos=[0, 0, -3] [wall: static, inverse_mass
      =0]
obj6: torus ring=1.5 tube=0.5, mass=1, charge=0, pos=[0, 0, 0]
obj7: point, mass=1, charge=0, pos=[1.40659, -0.995859, 0.569601]
obj8: sphere r=0.5, mass=2, charge=0, pos=[1.424704, 1.367496, -0.493612]
obj9: disk r=1, mass=0.6666666666666666, charge=0, pos=[-0.677386, -1.041493,
      -1.091679]
obj10: cuboid he=[0.5, 0.5, 0.5], mass=1.6666666666666667, charge=0, pos=[1.074397,
      0.816223, 1.102099]
obj11: cylinder r=0.5 h=1.5, mass=2, charge=0, pos=[-1.027024, -1.40389, -0.485619]
In[10]:= collide
Out[10]= collisions ON (51 collidable pair(s); 0 impulse(s) so far)
In[11]:= get obj0.inverse_mass
Out[11]= 0
In[12]:= get obj6.inertia_tensor
Out[12]= [[1.28125, 0, 0], [0, 1.28125, 0], [0, 0, 2.4375]]
In[13]:= energy
Out[13]= 30000
In[14]:= momentum
Out[14]= [100, 200, 100]
In[15]:= run 0.1 steps 100
Out[15]= t = 0.1 (2121 solver steps, 100 snapshots, |dE/E| = 2.040e-10, 119 collision(s)
      ) -- CONTACTS lists them)
In[16]:= energy
Out[16]= 29999.999993880127
In[17]:= momentum
Out[17]= [146.48911126803657, 102.1478121131382, 46.01601291636828]
In[18]:= get system.collisions
Out[18]= 119
In[19]:= get system.box
Out[19]= 4

```

*What to notice.* Cell 3 sizes the torus by the order-independent `inner_radius` + `outer_radius` pair; cell 8 uses `height` = 3/2 — the *full* height, so `half_height` reads 0.75. COLLIDE counts **51 collidable pairs**: 12 objects make 66 pairs, minus the 15 wall–wall pairs (two static bodies can never collide). Out[12] is the torus’s analytic inertia diag(1.28125, 1.28125, 2.4375) — exactly  $I_{xy} = m(\frac{1}{2}c^2 + \frac{5}{8}a^2)$ ,  $I_z = m(c^2 + \frac{3}{4}a^2)$  for  $c = 1.5$ ,  $a = 0.5$ . The energy starts at  $E_0 = \frac{1}{2} \cdot 1 \cdot |v|^2 = 30000$  **exactly** and, after **119 collisions** in 0.1s, is conserved to  $|dE/E| = 2.040 \times 10^{-10}$  — but momentum went from (100, 200, 100) to (146.49, 102.15, 46.02): **not conserved**, because the infinitely massive walls absorb momentum without moving — the physical signature of infinite mass. Along the way the point particle can *thread the torus hole* and passes through the ideal

zero-thickness disk (a measure-zero contact) — the ball-vs-anything collision tier is exact SDF geometry, not an approximation — while every finite-size body bounces off both. The walls end the run bit-identically at rest.

## 9.14 Example 14 — two dumbbells from a user-defined function

The two-release finale: a user function (§5.9) that builds **named dumbbells** (§5.1), called twice to launch a pair of tumbling dumbbells at each other, off-center and spinning. No walls this time — so *all three* conserved quantities must survive the collisions. The script is `scripts/collisions/12_two_dumbbells.p`. this transcript replays it.

```
In[1]:= set system.g_constant = 0
In[2]:= def create_dumbell(name, m1 = 1, m2 = 1, m_rod = 0.5, r1 = 0.25, r2 = 0.25,
    rod_radius = 0.1, length = 1, position = [0, 0, 0], velocity = [0, 0, 0],
    angular_velocity = [0, 0, 0]) {
    new dumbbell as name { m1 = m1, m2 = m2, m_rod = m_rod, r1 = r1, r2 = r2, rod_radius
    = rod_radius, length = length, position = position, velocity = velocity,
    angular_velocity = angular_velocity }
}
Out[2]= function create_dumbell(11 parameter(s)) defined -- 1 body line(s)
In[3]:= create_dumbell("dumbell0", 1, 2, 0.5, 0.25, 0.25, 0.1, 1, [-2, 0.15, 0], [1.5,
    0, 0], [0, 0, 0.6])
Out[3]= obj0 as dumbell0
In[4]:= create_dumbell("dumbell1", 2, 1, 0.4, 0.3, 0.2, 0.08, 1.2, [2, -0.15, 0],
    [-1.5, 0, 0], [0.4, 0, 0])
Out[4]= obj1 as dumbell1
In[5]:= list
Out[5]= obj0: dumbbell r1=0.25 r2=0.25 rod_r=0.1 len=1, mass=3.5, charge=0, pos=[-2,
    0.15, 0]
obj1: dumbbell r1=0.3 r2=0.2 rod_r=0.08 len=1.2, mass=3.4, charge=0, pos=[2, -0.15, 0]
In[6]:= get dumbell0.m1
Out[6]= 1
In[7]:= get dumbell1.m_rod
Out[7]= 0.39999999999999999
In[8]:= get dumbell0.vx
Out[8]= 1.5
In[9]:= energy
Out[9]= 7.865310611764706
In[10]:= momentum
Out[10]= [0.150000000000000036, 0, 0]
In[11]:= angmom
Out[11]= [0.4443030588235295, 0, -1.5059999999999998]
In[12]:= run 3 steps 60
Out[12]= t = 3 (3861 solver steps, 60 snapshots, |dE/E| = 9.463e-11, 2 collision(s) --
    CONTACTS lists them)
In[13]:= energy
Out[13]= 7.865310611020375
In[14]:= momentum
Out[14]= [0.150000000000000036, 0, 0]
In[15]:= angmom
Out[15]= [0.44430305882373156, -0.0000000000000009547918011776346, -1.505999999999855]
In[16]:= get system.collisions
Out[16]= 2
```

*What to notice.* Cell 2 is one DEF whose single body line does everything: `new dumbbell as name { ... }` forwards all eleven parameters, and because `name` arrives as a string ("dumbell10", "dumbell11"), each call registers the name it was given — `Out[3]/Out[4]` show both handles, `LIST` shows the part geometry, and cells 6–8 read the parts and the `.vx` shorthand through the names. (`Out[7]` reads 0.4 back as 0.3999999999999999: the parts are stored as mass *fractions* of the total, so one reconstruction rounding step shows through — 15 correct digits.) Then the physics: two tumbling, asymmetric rigid bodies meet off-center, collide **twice** (cell 16), and every conserved quantity survives the real CVODE event handling. Energy: 7.865310611764706 → 7.865310611020375 — the run banner prints  $|dE/E| = 9.463e-11$ . Momentum: [0.150000000000000036, 0, 0] **bit-identical**. Angular momentum about the origin — the delicate one, orbital  $r \times p$  plus spin, exchanged between the two at every off-center impact: conserved to  $\sim 10^{-13}$  per component (`Out[11]` vs. `Out[15]`), because the part-wise exact narrow phase applies the action–reaction impulse pair at one shared contact point. Contrast Example 13: there the infinitely massive walls absorbed momentum; here, with no walls,  $E$ ,  $P$  and  $L$  all hold at solver precision.

## 9.15 Example 15 — a particle in a box, solved inside the language

The special functions are not decoration: with `eigenvalues` you can set up and solve a small quantum problem in the notebook itself.

Take a particle in an infinite square well of width  $L = 1$  with  $\hbar = m = 1$ . Discretise  $-\frac{1}{2}\psi'' = E\psi$  on 4 interior points, spacing  $h = 0.2$ . The second-derivative stencil puts  $1/h^2$  on the diagonal and  $-1/2h^2$  beside it, so the Hamiltonian is a  $4 \times 4$  tridiagonal matrix you can type out. The exact answer is  $E_n = n^2\pi^2/2$ , so  $E_1 = \pi^2/2$ .

```
In[1]:= let h = 0.2
Out[1]= h set
In[2]:= let k = 1 / (h * h)
Out[2]= k set
In[3]:= eigenvalues([[k, -0.5*k, 0, 0], [-0.5*k, k, -0.5*k, 0], [0, -0.5*k, k, -0.5*k],
                    [0, 0, -0.5*k, k]])
Out[3]= [4.774575140626312, 17.274575140626325, 32.72542485937371, 45.225424859373675]
In[4]:= pi * pi / 2
Out[4]= 4.934802200544679
```

The ground state comes out at 4.7746 against an exact 4.9348 — **3.2% low**. That is not a bug, and it is worth understanding rather than tightening a tolerance until it goes away. A 3-point stencil on 4 points is a *coarse* grid, and its eigenvalues have the closed form  $k(1 - \cos(n\pi/5))$  with  $k = 1/h^2 = 25$  — which reproduces all four numbers above exactly ( $25(1 - \cos 36^\circ) = 4.7746$ ). The discretisation error is real, known, and second order in  $h$ : refine the grid and it falls off as  $h^2$ .

Note the row shapes. Each row here has four entries, so it lexes as a *quaternion* rather than a list — the bracket literal is overloaded (see §4.1). It works anyway, because every bracket shape is accepted where a numeric list is wanted. You should not have to think about quaternions to type a matrix.

The rest of the family is there to be checked against its own identities, which is the honest way to use a special-function library you did not write:

```
In[5]:= let t = 0.7
Out[5]= t set
```



```

In[6]:= chebyshev_t(5, cos(t)) - cos(5 * t)
Out[6]= -0.00000000000000011102230246251565
In[7]:= sph_j(0, 1.3) - sin(1.3) / 1.3
Out[7]= 0
In[8]:= legendre_p(3, 0.4) - 0.5 * (5 * 0.4 * 0.4 * 0.4 - 3 * 0.4)
Out[8]= 0.00000000000000011102230246251565
In[9]:= bessej_array(4, 2)
Out[9]= [0.22389077914123567, 0.5767248077568734, 0.35283402861563773,
0.12894324947440208, 0.03399571980756844]

```

In[6] is  $T_n(\cos \theta) = \cos n\theta$ ; In[7] is  $j_0(x) = \sin x/x$ ; In[8] is  $P_3(x) = (5x^3 - 3x)/2$ . All three land at zero or one part in  $10^{16}$  — a single rounding step. In[9] returns the whole table  $J_0 \dots J_4$  in one call, which is what a Bessel-expanded propagator actually needs.

Finally, the thing the library refuses to do:

```

In[10]:= hermite_h(2.5, 1)
Err[10]: hermite_h(): argument 1 must be a whole number (an integer order), got 2.5

```

There is no Hermite polynomial of order 2.5. The alternative — silently computing  $H_2$  — would hand you a plausible number with no indication anything went wrong, and you would carry it through the rest of the calculation.

## 9.16 Example 16 — quantum mechanics: bound states, then tunnelling

The QM family solves an actual quantum problem in the notebook. Start with the harmonic oscillator, whose answer everyone knows: with  $\hbar = m = \omega = 1$  the spectrum is  $E_n = n + \frac{1}{2}$ . The potential is an ordinary user function.

```

In[1]:= def v(x) { 0.5 * x * x }
Out[1]= function v(1 parameter(s)) defined -- 1 body line(s)
In[2]:= qm grid -8 8 250
Out[2]= grid [-8, 8] with 250 interior points, h = 0.063745 (potential and psi cleared)
In[3]:= qm potential v
Out[3]= potential 'v' sampled at 250 points, V in [0.000507928445580228,
31.49207155441977] (psi cleared)
In[4]:= qm states 4
Out[4]= 4 lowest bound state(s):
E[0] = 0.499872985615
E[1] = 1.499364798834
E[2] = 2.498348101796
E[3] = 3.496822505539

```

0.4999, 1.4994, 2.4983, 3.4968 against an exact 0.5, 1.5, 2.5, 3.5. The error grows with  $n$  — that is not noise, it is the three-point stencil: higher states oscillate faster and a coarse grid resolves them less well, with the error tracking  $2n^2 + 2n + 1$ .

Now the sharpest test available. A stationary state is *stationary*, so loading one and propagating it must change nothing:

```

In[5]:= qm state 1
Out[5]= psi = bound state 1, E = 1.499364798834, t reset to 0

```

```

In[6]:= qm energy
Out[6]= 1.499364798833686
In[7]:= qm run 4 steps 400
Out[7]= t = 4 (400 step(s) of dt = 0.01), <E> = 1.499364798834, norm drift = 4.219e-15
In[8]:= qm energy
Out[8]= 1.499364798833692

```

Four hundred Crank–Nicolson steps moved the energy by  $6 \times 10^{-15}$  — one bit — and the norm by  $4 \times 10^{-15}$ . The Cayley operator is unitary for *any* time step, so that drift measures the linear solver rather than  $dt$ ; and the energy holding still couples the eigensolver to the propagator, so it would break if either were wrong or if they disagreed about the Hamiltonian.

Now tunnelling, the problem with no classical analogue. A packet of central energy  $E_0 = k_0^2/2 = 2$  is fired at a barrier of height 2.5 — *above* its energy, so classically nothing gets through:

```

In[1]:= qm grid -100 100 2000
Out[1]= grid [-100, 100] with 2000 interior points, h = 0.099950 (potential and psi
        cleared)
In[2]:= qm potential barrier 2.5 0 1
Out[2]= potential 'barrier 2.5 on [0, 1]' sampled at 2000 points, V in [0, 2.5] (psi
        cleared)
In[3]:= qm packet -25 2 2
Out[3]= psi = Gaussian packet at x0 = -25, sigma = 2, k0 = 2, t = 0
In[4]:= qm run 30 steps 3000
Out[4]= t = 30 (3000 step(s) of dt = 0.01), <E> = 2.023971780446, norm drift = 3.018e
        -13
In[5]:= qm prob 1 100
Out[5]= 0.327563019391693
In[6]:= qm prob -100 0
Out[6]= 0.672436408645103

```

**33 % of the particle got through a barrier it did not have the energy to cross.** The two channels add to 1.000000 — nothing was lost, and nothing reached a wall (there was no warning). The domain is 200 wide precisely so the reflected packet has nowhere to bounce from within  $t = 30$ .

Note the grid sizes. Bound states used 250 points because that calculation diagonalises a dense matrix and costs  $O(n^3)$ ; scattering used 2000 because propagation is  $O(n)$  per step and can afford the resolution. Choosing both to be the same would waste one or starve the other.

### 9.17 Example 17 — tunnelling, with the barrier as a user function

Comparison operators exist so that a piecewise potential can be an ordinary function. This is the canonical 1-D quantum problem written that way, and it is the notebook `dynamic_notebooks/tunneling.posim`.

```

In[1]:= def barrier(x) { 2.5 * (x > 0) * (x < 1) }
Out[1]= function barrier(1 parameter(s)) defined -- 1 body line(s)
In[2]:= barrier(-1)
Out[2]= 0
In[3]:= barrier(0.5)
Out[3]= 2.5
In[4]:= barrier(2)
Out[4]= 0

```

$(x > 0) * (x < 1)$  is 1 inside the interval and 0 outside — an indicator function built from arithmetic on truth values.

A packet of central energy  $E_0 = k_0^2/2 = 2$  is fired at a barrier of height 2.5, *above* its energy:

```
In[5]:= qm grid -100 100 2000
Out[5]= grid [-100, 100] with 2000 interior points, h = 0.099950 (potential and psi
        cleared)
In[6]:= qm potential barrier
Out[6]= potential 'barrier' sampled at 2000 points, V in [0, 2.5] (psi cleared)
In[7]:= qm packet -25 2 2
Out[7]= psi = Gaussian packet at x0 = -25, sigma = 2, k0 = 2, t = 0
In[10]:= qm run 30 steps 3000
Out[10]= t = 30 (3000 step(s) of dt = 0.01), <E> = 2.023971780446, norm drift = 3.018e
        -13
In[11]:= qm prob 1 100
Out[11]= 0.327563019391693
In[12]:= qm prob -100 0
Out[12]= 0.672436408645103
```

**33 % crossed a barrier it had no classical right to cross**, and the two channels account for all but  $10^{-6}$  of the probability.

Note that `qm potential barrier` picked the *user's* function, not the built-in shape of the same name. A bare name is always yours; the built-in needs arguments.

### Watching it happen

```
In[14]:= qm animate "scatter.html" 32 frames 140
Out[14]= wrote scatter.html -- 140 frames over t = 32 (dt = 0.011429, 667 points per
        frame), worst norm drift 5.822e-13. Open it in a browser.
```

A self-contained page — nothing fetched from the network — showing  $|\psi|^2$  against  $x$  with the potential overlaid, play/pause, a scrub bar, and a live norm and transmitted readout computed in the browser. Its final transmitted value is 0.327562 against the notebook's 0.327563: two independent implementations of the same integral.

### The same answer on half the domain

```
In[15]:= qm grid -45 45 1350
In[17]:= qm absorb 15 3
Out[17]= absorbing edges: width 15, strength 3, power 2. Propagation is NO LONGER
        unitary.
In[19]:= qm run 20 steps 2000
Out[19]= t = 20 (2000 step(s) of dt = 0.01), <E> = 2.027672292154, norm drift = 1.056e
        -4
In[20]:= qm prob 1 30
Out[20]= 0.327689667523621
```

0.327690 against 0.327563 — four parts in ten thousand, on a domain **less than half the size**.

## A double barrier, and an honest negative result

```
In[24]:= def double(x) { 2.5 * ((x > 0) * (x < 1) + (x > 3) * (x < 4)) }  
In[31]:= qm prob 4 100  
Out[31]= 0.271498902829932
```

Two barriers with a gap form a resonant cavity. At this energy it transmits **0.2715**, less than the single barrier's **0.3276**, with about 0.8 % of the probability left in the gap. That trapped fraction *is* the cavity; but resonant enhancement occurs only at its quasi-bound energies, and those peaks are narrower than this packet's momentum spread ( $\sigma = 2$  gives  $\Delta k = 0.25$ ), which averages straight over them. A scan over  $k_0$  from 1.4 to 2.9 rises monotonically with no peak. See TUNNELING\_RESULTS.md.

## 9.18 Example 18 — the double slit, in two dimensions

The canonical 2-D quantum problem, and the notebook `dynamic_notebooks/double_slit.posim`. The wall is an ordinary user function of two arguments, built entirely from comparisons:

```
In[1]:= def openings(y) { (y > 1.5) * (y < 2.5) + (y > -2.5) * (y < -1.5) }  
In[2]:= def slit(x, y) { 60 * (x > 0) * (x < 0.4) * (1 - openings(y)) }  
In[3]:= slit(0.2, 0)  
Out[3]= 60  
In[4]:= slit(0.2, 2)  
Out[4]= 0
```

```
In[8]:= qm2 grid -12 24 450, -16 16 400  
In[10]:= qm2 absorb 4, 8  
In[11]:= qm2 packet -4.5, 0, 1.2 4, 8 0  
In[12]:= qm2 energy  
Out[12]= 31.008336923103123  
In[15]:= qm2 run 2.6 steps 800  
Out[15]= t = 2.6 (800 ADI step(s) of dt = 0.00325), <E> = 30.990650438623, norm drift =  
7.796e-1  
In[18]:= qm2 norm  
Out[18]= 0.220410865359174
```

Those three lines together say something worth pausing on: **78 % of the probability was absorbed at the walls, and the energy moved by 0.06 %**. The absorber removed the reflected wave — the wall is 60 high against a packet of energy 32, so most of it bounces — without touching the energy of what remains.  $\langle E \rangle$  is  $\langle \psi | H | \psi \rangle / \langle \psi | \psi \rangle$ , divided by the norm on purpose; undivided it would have fallen with the norm and looked like an energy leak.

The fringes, counted in bands on a screen at  $11 < x < 19$ :

| band        | P              | expected                   |
|-------------|----------------|----------------------------|
| $0 \pm 0.5$ | <b>0.01577</b> | central maximum            |
| 0.5–1.5     | 0.00452        | first minimum, $y = 1.46$  |
| 2.5–3.5     | <b>0.01742</b> | first maximum, $y = 2.96$  |
| 3.5–4.5     | 0.00471        | second minimum, $y = 4.56$ |
| 5.5–6.5     | <b>0.01399</b> | second maximum, $y = 6.32$ |

With slit separation  $d = 4$  and  $\lambda = 2\pi/k = 0.785$ , maxima sit at  $\sin\theta = m\lambda/d$  with the screen  $L \approx 14.8$  from the slits, giving 2.96 and 6.32. **Every predicted maximum and minimum lands in the right band.**

Two corrections got there. A first attempt used  $k = 4$ ,  $d = 2$ , putting the first maximum at  $52^\circ$  — off the screen, so the scan showed a featureless lobe. A second launched the packet *inside* the absorber, and 86 % of it was eaten before reaching the slits.

```
In[28]:= qm2 animate "double_slit.html" 3 frames 100
```

A self-contained heat-map page. Its own in-page analysis of the final frame finds fringe maxima at  $y = 0.04$  and  $y = 3.23$ , matching the band scan and the theory independently of the Rust that produced it.

## 10 Error message tour

| you type                           | you get                                                                                                                                        |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| get obj7.mass (no obj7)            | no object obj7                                                                                                                                 |
| get obj0.bogus                     | unknown object field 'bogus' -- see HELP for the field list                                                                                    |
| [1,0,0] * [0,1,0]                  | cannot multiply vec3 by vec3 (for vec3*vec3 use dot()/cross())                                                                                 |
| set = 3                            | parse error at column 5: expected a path root ('objN' or 'system'), found '='                                                                  |
| run 1 under SPRK with $B \neq 0$   | SPRK method requires a separable Hamiltonian: magnetic field B must be zero ... use METHOD ADAMS or BDF                                        |
| bogusname                          | unknown name 'bogusname' (define it with LET, pass it as a function parameter, or use 'bogusname.field' for a registered object)               |
| get d.m1 (only ball is registered) | no object named 'd' (registered names: ["ball"]; NEW ... AS <name> creates one; positional paths are objN.field, contactK.field, system.field) |
| new sphere as d (name taken)       | the name 'd' already refers to obj0 -- DEL it or pick another name                                                                             |
| new sphere as system               | 'system' is reserved                                                                                                                           |
| show nosuch                        | no user function 'nosuch' -- FUNCS lists the defined ones                                                                                      |

Every error names the column or the exact field/feature at fault, and never aborts the session.

## 11 The numerical engine underneath (SUNDIALS 7.8.0)

You never name a solver library when you type **STEP** or **RUN** — but it is worth one page knowing what answers you.

**SUNDIALS** is Lawrence Livermore National Laboratory's suite of differential-equation solvers,

in production use in physics codes since the 1990s. `sundials_rs/` in this repository is a *pure-Rust translation of SUNDIALS 7.8.0*: no `unsafe`, no FFI, no C compiler, no `crates.io` dependency. It is vendored byte-for-byte from `once-ere/SUNDIALS_7_8_Rust_port_for_Linux@780b916`; the upgrade from the 7.7.0 engine the predecessor repository used is recorded, with its evidence, in `PORT_7.8.0_PROVENANCE.md`.

### 11.1 Which solver runs which command

| you type                          | what runs                                                                                                                                 | from                       |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|----------------------------|
| METHOD ADAMS (default)            | CVODE, variable-order variable-step Adams–Moulton, Newton iteration, dense difference-quotient Jacobian                                   | <code>cvode_rs</code>      |
| METHOD BDF                        | CVODE BDF — the stiff-problem family                                                                                                      | <code>cvode_rs</code>      |
| METHOD SPRK <i>table [dt]</i>     | ARKODE SPRKStep, symplectic partitioned Runge–Kutta at a fixed step                                                                       | <code>arkode_rs</code>     |
| COLLIDE ON with a collidable pair | the same solver with <i>rootfinding</i> armed on the pairwise signed separations; a downward zero crossing <i>is</i> the moment of impact | <code>CVodeRootInit</code> |

There is deliberately no fourth option. Every trajectory this simulator prints came out of an error-controlled or symplectic solver; the legacy Euler and Verlet steppers were not carried over.

### 11.2 What “rootfinding” buys you

A naive collision check samples the geometry once per output frame and asks “are they overlapping now?”. If the ball moved farther than the wall is thick, the answer is no on both sides and the ball passes through. Rootfinding asks a different question: it hands the solver a continuous function — here, the signed gap between two surfaces — and the solver *interpolates back to the instant that function crossed zero*. The bounce happens at the exact time of impact, not at the next frame boundary.

```
In[1]:= new sphere { mass = 1, radius = 0.5, position = [0, 4.5, 0] }
In[2]:= new cuboid { mass = 1, half_extents = [10, 0.05, 10],
                  position = [0, -0.05, 0], inverse_mass = 0 }
In[3]:= set system.uniform_gravity = [0, -10, 0]
In[4]:= collide on
In[5]:= run 1 steps 100
Out[5]= t = 1 (209 solver steps, 100 snapshots, |dE/E| = 2.890e-11,
          1 collision(s) - CONTACTS lists them)
In[6]:= contacts
Out[6]= contact0: obj0 <-> obj1 at t = 0.89442719099999157
        point   = [0, 0, 0]
        normal  = [0, -1, 0] (from obj0 toward obj1)
        approach speed = 8.944271909999916, impulse = 17.88854381999832
```

Work out the answer by hand. The plate’s top face is at  $y = 0$  (centre  $-0.05$ , half-thickness  $0.05$ ); the ball’s centre starts at  $4.5$  and its radius is  $0.5$ , so the surfaces meet after the centre falls  $4.0$ . Under  $g = 10$  that is  $t = \sqrt{2 \cdot 4/10} = \sqrt{0.8}$ :

```
analytic  sqrt(0.8) = 0.89442719099999159
measured          = 0.89442719099999157
```

|                |            |
|----------------|------------|
| relative error | = 1.24e-16 |
|----------------|------------|

One part in  $10^{16}$  — a single unit in the last place of a 64-bit float. Nothing about the output cadence produced that: the run asked for 100 snapshots, and none of them falls anywhere near 0.894.... The solver interpolated to the crossing.

### 11.3 All six families, and what reaches each

| crate     | solves                                                       | reached by                    |
|-----------|--------------------------------------------------------------|-------------------------------|
| cvmode_rs | ODE initial-value problems (Adams / BDF)                     | METHOD ADAMS, METHOD BDF      |
| arkode_rs | Runge–Kutta: explicit, implicit, IMEX, multirate, symplectic | METHOD SPRK                   |
| ida_rs    | differential-algebraic systems<br>$F(t, y, \dot{y}) = 0$     | CONSTRAIN + METHOD IDA        |
| kinsol_rs | nonlinear algebraic systems, Anderson acceleration           | EQUILIBRIUM                   |
| cvodes_rs | CVODE plus forward and adjoint sensitivity analysis          | SENSITIVITY                   |
| idas_rs   | IDA plus sensitivity analysis                                | SENSITIVITY while constrained |

Every one of them is diffed against the upstream C reference output — see `sundials_rs/VERIFICATION.md`. The last four arrived with the 7.8.0 engine and were wired into the language afterwards; the shape of that wiring is in `ARCHITECTURE.md` §3.9.

## 12 Constrained dynamics, equilibrium and sensitivity

Everything so far answers “*what happens next?*”. Three more commands ask different questions, each answered by a different solver.

| you type                     | question                                     | solver         |
|------------------------------|----------------------------------------------|----------------|
| STEP / RUN                   | what happens next?                           | CVODE / ARKODE |
| CONSTRAIN + METHOD IDA + RUN | ...with this geometry held exactly           | IDA            |
| EQUILIBRIUM                  | where does it come to rest?                  | KINSOL         |
| SENSITIVITY                  | how much does the answer depend on an input? | CVODES / IDAS  |

### 12.1 The four joints

| command                    | rows | holds                             | freedoms left |
|----------------------------|------|-----------------------------------|---------------|
| CONSTRAIN <i>a b [len]</i> | 1    | a fixed distance                  | 5             |
| BALL <i>a b</i>            | 3    | a shared point                    | 3             |
| UNIVERSAL <i>a b u w</i>   | 4    | a shared point, two shafts square | 2             |
| HINGE <i>a b axis</i>      | 5    | a shared point and an axis        | 1 (the swing) |

The last three grip *orientation*, which is why METHOD IDA carries the full 13-numbers-per-object state rather than positions alone. **The pivot is the midpoint of the two bodies as they stand when you make the joint**, carried into each body’s frame — the same “freeze what you have” rule a bare CONSTRAIN follows, so the joint is satisfied the instant it is made.

A door is an anchor, a slab and a hinge:

```
In[4]:= new sphere as jamb { mass = 1, radius = 0.02,
                             position = [0, 0, 0], inverse_mass = 0 }
In[5]:= new cuboid as door { mass = 1, half_extents = [0.2, 0.4, 0.2],
                             position = [0.0199986666933331, -0.9998000066665778, 0] }
In[6]:= hinge jamb door [0, 0, 1]
In[9]:= run 1 steps 10
Out[9]= t = 1 (70 solver steps, 10 snapshots, |dE/E| = 1.613e-9)
In[10]:= constraints
Out[10]= constraint0: hinge obj0 <-> obj1, 5 row(s)
worst |g| = 2.73750133672479e-10, worst |g_dot| = 2.3769240437118873e-9
In[11]:= get door.angular_momentum
Out[11]= [0, 0, 0.003742219622967182]
```

The door swings about  $z$  and *nothing else* — the two rows a hinge has over a ball joint are exactly the ones that forbid the other two axes — and the joint is held to  $2.7 \times 10^{-10}$  after 70 steps. A hinged rigid body is a *compound* pendulum, with period  $T = 2\pi \sqrt{I_{\text{pivot}}/(mgd)}$  and  $I_{\text{pivot}} = I_{\text{com}} + md^2$ ; the simulator reproduces it to about  $3 \times 10^{-8}$  of a full swing.

**A joint constrains velocity, not just position.** A ball joint says the two bodies share a point, so at the velocity level it says  $v_i + \omega_i \times r_i = v_j + \omega_j \times r_j$  — a body turning about a pivot offset from its centre *must have its centre moving*. Give it a spin and leave its velocity at zero and the state is not on the constraint manifold at all. The run **projects** the starting velocities onto it — the smallest mass-weighted change that satisfies the joint, which is exactly the impulse a real coupling delivers when clutched onto a spinning shaft — and reports how big that change was. An already-consistent state is left exactly alone. (A rod has no angular Jacobian, so spin never enters its  $\dot{g}$ ; that is why rods carried spinning bodies from the start and hid this.)

**One real limit.** Orientation joints carry a tolerance floor of  $\text{rtol} = 10^{-6}$ , because the differential-algebraic system a hinge produces is *index 2* and has an accuracy ceiling no tolerance can push past — measured across twelve compound pendulums,  $10^{-6}$  converges in every one and  $10^{-8}$  in none. RUN reports when the floor was applied.

BALL, HINGE and UNIVERSAL are *contextual* keywords — commands only at the start of a line — so new sphere as ball and get ball.mass keep working.

## 12.2 CONSTRAIN — a rod, not a spring

|                            |                                       |
|----------------------------|---------------------------------------|
| CONSTRAIN <a> <b> [length] | rigid rod between two objects         |
| CONSTRAIN OFF              | drop every rod                        |
| CONSTRAINTS                | list them, and how well they are held |

<a> and <b> are objects, written either positionally (obj0) or by a name registered with NEW ... AS. **With no length the rod freezes the separation the two bodies already have**, which is always consistent and is what you almost always want.



You could model a rod as a very stiff spring. That is not the same thing: a stiff spring is an approximation that vibrates, needs a tiny step, and still lets the length wander. A constraint is an *algebraic equation* the motion must satisfy exactly. Adding one changes the problem from an ODE into a **differential-algebraic equation**, and a constrained system refuses every method but IDA, by name:

```
In[7]:= run 1 steps 5
Err[7]: this system has 1 rigid constraint(s), which only the DAE
        integrator can hold: use METHOD IDA (or remove them with
        CONSTRAIN OFF). The current method is Adams
```

**A body with `inverse_mass = 0` is an anchor.** It never moves and it absorbs the rod's reaction. Anchor + bob + rod is a pendulum.

**What is held, and how well.** CONSTRAINTS reports the worst  $|g|$  (how far the rod is from its length) and  $|g'|$  (how fast that is changing). Both stay at roundoff for the whole run, because the formulation carries them *both* as equations. A cheaper scheme that constrains only the acceleration lets  $g$  drift quadratically, and by the time you notice, the answer is quietly wrong.

**Scope.** Constraints act on positions. A spinning rigid body, or an external torque, is refused with a message naming it.

### 12.3 EQUILIBRIUM — where does it come to rest?

Finds a configuration where every free body has zero net force, every anchor is where it started, and every rod is the right length; moves the bodies there and stops them. `system.time` is untouched — this is not an integration.

```
In[7]:= equilibrium
Out[7]= equilibrium found in 17 Newton iteration(s), 67 residual
        evaluation(s); largest net force on any free body =
        7.459152323898993e-13, worst |g| = 1.9317880628477724e-14
```

The starting guess is wherever the bodies already are, so it finds *the nearby* rest state. **It says nothing about stability** — a pencil balanced on its point is an equilibrium. Perturb the answer and RUN if you need to know. And a system where every body is free has no isolated equilibrium at all (translate it and nothing changes); the refusal says exactly that.

### 12.4 SENSITIVITY — how much does the answer depend on the input?

```
SENSITIVITY <t> "<param>" ["<param>" ...]
```

Runs for `<t>` and reports  $\partial(\text{state})/\partial p$  for each parameter. Names are quoted strings because `mass 0` is two tokens: `"g_constant"`, `"mass 0"`, `"charge 1"`, `"gravity.y"`, `"e_field.x"`, `"b_field.z"`.

Running twice with slightly different inputs and subtracting is the difference of two nearly equal numbers and loses most of its digits. This integrates the derivative *alongside* the state instead. CVODES normally; IDAS when the system is constrained, because only then are the equations a DAE.

Free fall gives  $y(T) = \frac{1}{2}gT^2$ , so  $\partial y/\partial g = T^2/2$ , which at  $T = 3$  is exactly 4.5:

```
In[15]:= sensitivity 3 "gravity.y" "mass 0"
Out[15]= t = 3 (CVODES, 129 solver steps)
d/d(gravity.y):
  obj0 position [0, 4.500000056696235, 0]
d/d(mass 0):
  obj0 position [0, 0, 0]
```

The second answer is worth a moment. In uniform gravity every mass accelerates equally, so the trajectory does not depend on the mass **at all** — and the derivative comes back as exactly zero, not as a small number. That is the difference between a real sensitivity calculation and a finite difference.

## 12.5 The one rule that changed for anyone reading the Rust

The 7.8.0 translation models C’s opaque pointers directly. An `N_Vector` is a handle, and its payload is reached through `N_VGetArrayPointer`, which returns a *borrow guard*:

```
let y = N_VNew_Serial(neq, &sunctx).ok_or("N_VNew_Serial returned NULL"?);
{
    let mut d = N_VGetArrayPointer(&y).expect("serial vector");
    d[0] = 1.0;
}
// <- guard dropped here, ON PURPOSE
CNode(&cnode_mem, tout, &y, &mut t, CV_NORMAL);
```

Holding that guard across the `CNode` call would be a live borrow of a vector the solver is about to write. `physical_object/src/integrate.rs` therefore wraps every access in `with_data / with_data_mut`, which take the guard, do the work and drop it. Nothing about the command language changes.

## 13 Browser videos: watching a run after it has finished

SCENE CREATE (§5.6) gives you a *live* window: it needs a running posim behind it. The other thing you often want is a file — something you can mail to a colleague, open next year on a laptop with no Rust toolchain, and still scrub frame by frame.

`tools/record_video.py` makes one:

```
cargo build --release -p posim # once
recorder/src/record_video.py videos/scenes/kepler_ellipse.posim \
  -o videos/kepler_ellipse.html \
  --frames 360 --dt 0.02 \
  --title "Kepler orbit, e = 0.6" \
  --caption "CVODE Adams, 360 frames of dt = 0.02."
```

### 13.1 How it works, and what it does not do

1. It starts `posim -machine` — the same JSON protocol JupyterLab uses (§7).
2. It runs your setup script one line at a time.

3. Then it loops: ask for `{"op":"state"}`, keep the answer, send `step <dt>`, repeat.
4. It writes one HTML file with those frames embedded and a plain canvas player around them.

**Every advance in step 3 is a real SUNDIALS step.** The tool has no integrator of its own; it is a camera, not a physics engine. And the page it writes fetches nothing: no CDN, no font server, no analytics. Open it with `file://` on a machine with no network and it works.

## 13.2 The player

| control                    | does                                                  |
|----------------------------|-------------------------------------------------------|
| Play / Pause, or Space     | run the recording                                     |
| ◀◀ / ▶▶, or ← →            | one frame back / forward                              |
| scrub bar                  | jump anywhere                                         |
| speed                      | 0.25× to 4×                                           |
| drag                       | orbit the camera                                      |
| wheel, or + / −            | zoom                                                  |
| ↑ ↓                        | pan                                                   |
| Reset view                 | back to the framing it opened with                    |
| trails / labels / contacts | motion trails, body names, gold contact-normal arrows |

The readout in the corner is live per frame: the frame number,  $t$ , the total energy,  $|P|$ ,  $|L|$ , the running collision count, and the method and  $dt$  the recording was made with. **This is the point of the format:** you can stop on the frame where something looks wrong and read the conserved quantities off it.

### 13.3 The seven shipped recordings

| file                           | what to watch for                                                                                                                                                    | measured                                                                                                                                           |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| videos/kepler_ellipse.html     | the speed swinging between perihelion and aphelion on an $e = 0.6$ ellipse                                                                                           | $ dE /E = 9.8\text{e-}8$ ,<br>$ dL / L  = 1.3\text{e-}7$                                                                                           |
| videos/tumbling_rackety.html   | the Dzhanibekov flip: a torque-free cuboid spun about its <i>intermediate</i> axis turns over, and over                                                              | $ dL / L  = 0$ <b>exactly</b> ;<br>$ dE /E = 6.4\text{e-}9$                                                                                        |
| videos/box_of_shapes.html      | a cylinder, a disk and a cuboid rattling in a rigid BOX 4; gold arrows are the analytic contact normals, sized by impulse                                            | 36 collision events,<br>$ dE /E = 3.4\text{e-}16$                                                                                                  |
| videos/double_pendulum.html    | two HINGE joints assembled into the chaotic linkage; the gold rings are the joints                                                                                   | the joints hold to $ g  = 5.6\text{e-}8$ ; energy wanders 3 parts in 10,000                                                                        |
| videos/universal_joint.html    | a UNIVERSAL joint carrying a driven shaft's rotation to a second shaft; the bend flattens out straight and folds back, and the speed across the joint swings with it | the bend stops at $\cos \beta = 0.6000004$ against a geometric bound of exactly 0.6; the three joints hold to $ g  = 4.0\text{e-}7$                |
| videos/ball_joint_chain.html   | four BALL joints, whirling as they collapse; the chain leaves the plane it started on, which a hinged chain cannot                                                   | the four joints hold to $ g  = 3.3\text{e-}9$ ; $ z $ runs from exactly 0 to 1.7147                                                                |
| videos/rod_pendulum_chain.html | four CONSTRAIN rods, the cheapest linkage there is at one row each, going chaotic                                                                                    | run continuously at the default tolerance the rods hold to $ g  = 5.4\text{e-}15$ ; this recording, 250 cold restarts, holds $ g  = 7.8\text{e-}8$ |

### 13.4 A trap worth knowing about

The box recording sets `system.g_constant = 0` on its first line, and that line is not decoration. posim's default  $G$  is 1, so three bodies rattling inside a small box also *attract each other*, and the softened pairwise force at `softening = 1e-6` is very nearly singular when two surfaces touch. With gravity left on, the same scenario drifts **3.2 %** in energy through 25 events; with  $G = 0$  it holds to  $5.1\text{e-}16$  through 36. Neither number is a solver defect — they are different physical systems.

The lesson generalizes: **a conservation claim carries its system settings with it**. When a run's  $|dE|/E$  surprises you, check `ENERGY`, `system.g_constant`, `system.softening` and `system.uniform_gravity` before suspecting the integrator.

### 13.5 What a BALL joint buys, measured exactly

BALL and HINGE are easy to describe and easy to confuse: both hold a point, and a hinge additionally holds an axis. The ball-chain recording turns that sentence into a number.

Take four links laid end to end and start them as a **rigid rotation** of the whole assembly about

the vertical,  $v = \omega \times r$  with  $\omega = [0, 1.5, 0]$ , so  $v = [0, 0, -1.5x]$ . A rigid motion moves nothing relative to anything, so it violates no joint of any kind — as a *position* constraint. The velocity constraint is where the two joints part company:

| the same four links, joined by | worst $ g $ | worst $ \dot{g} $ |
|--------------------------------|-------------|-------------------|
| four BALL joints               | 0 exactly   | <b>0 exactly</b>  |
| four HINGE joints about $z$    | 0 exactly   | <b>1.5</b>        |

1.5 is not approximately anything. It is  $\Omega$ , the whirl rate, because the whirl is precisely the component a hinge about  $z$  forbids, and the velocity residual of a forbidden motion is its own magnitude.

**What follows from it.** A start with  $|\dot{g}| \neq 0$  is not on the constraint manifold, so before integrating anything the solver must project the velocities onto it, and that projection *changes the motion you asked for*. The hinge chain cannot be given this whirl; it can only be given whatever remains after the whirl is removed. The ball chain takes it untouched, and `initial_velocity_projected` is 0.

Watch the recording's  $z$  axis for the consequence. Every link starts exactly on the plane  $z = 0$ , and the chain reaches  $|z| = 1.7147$ . A hinged chain would still be at exactly zero, for ever, because that is what fixing an axis means.

### 13.6 A restart is not a continuation

The rod-chain recording exists to document a trap that costs an afternoon if you meet it without warning.

CONSTRAIN is the best-conditioned joint in the language — one well-scaled scalar equation,  $g = |d| - L$ , with none of the index-2 orientation coupling that forces a tolerance floor on BALL, HINGE and UNIVERSAL. So a rod-only system is correctly *not* floored, and runs at whatever you asked for. Four rods, five seconds, default `rtol` =  $10^{-10}$ , `atol` =  $10^{-12}$ , give a worst  $|g|$  of  $5.4 \times 10^{-15}$ : roundoff. Record the same chain and it fails on the second frame, with `IDASolve` returning  $-4$ .

Nothing about the physics changed. What changed is that **a recording is not one integration**. It is one STEP per frame, and every STEP is a *cold restart* — a fresh solver, a fresh multiplier seed, no BDF history to lean on. The tolerance a continuous run sustains is not the tolerance a restart sustains, and for this chain the gap is four orders of magnitude:

| four rods, five seconds     | continuous RUN              | 250 STEPs                  |
|-----------------------------|-----------------------------|----------------------------|
| default $10^{-10}/10^{-12}$ | $ g  = 5.4 \times 10^{-15}$ | fails on restart 2         |
| floor $10^{-6}/10^{-8}$     | fine                        | $ g  = 7.8 \times 10^{-8}$ |

BALL, HINGE and UNIVERSAL never meet this, and now the reason is visible: they are floored to exactly  $10^{-6}/10^{-8}$  *automatically*, so every orientation-joint recording has been asking for restart-safe tolerances without knowing it. The floor those joints get for *accuracy* is also what makes them recordable frame by frame. A rod-only scene has to ask by hand.

**How to recognize it.** A RUN that succeeds and a loop of STEPs that fails, at the same tolerance on the same system, is this and not a physics bug. Loosen the tolerance, or do not chop the integration up.

**A related sharp edge.** This chain is chaotic, and its margin at the default tolerance is thin enough to be moved by *rounding the initial positions*. Written to six decimals, a five-rod version survives the continuous run; written exactly, it fails at  $t = 0.43$ . Four rods is robust either way,

which is why the scene ships with four and writes its coordinates out in full. If a constrained run's success depends on how you rounded the input, you are on a knife edge, not on a result.

### 13.7 Count your rows before you brace a mechanism

The universal-joint recording is worth reading as a design exercise, because the obvious way to build it does not work.

A **UNIVERSAL** holds two things: one shared point, and one right angle between the two trunnion axes. It does *not* hold the two shafts straight. The bend between them is free — that is the whole reason the joint exists — so if the output shaft is left dangling it does not transmit anything, it just swings down past the joint like a pendulum and folds back on the input shaft at  $176^\circ$ .

So the output shaft has to be braced. The textbook drawing puts it in a second bearing, and a bearing is a **HINGE**. Try it and IDA quits at  $t = 0$  with `retval = -4`. Count the rows and the reason is arithmetic, not numerics. Two free shafts are 12 freedoms, and

$$\underbrace{5}_{\text{HINGE}} + \underbrace{4}_{\text{UNIVERSAL}} + \underbrace{5}_{\text{HINGE}} = 14 \text{ rows on 12 freedoms.}$$

Three of those rows are *redundant*: once both shafts are held in bearings whose axes meet, the shared point the universal joint asks for is already implied by the geometry. Redundant rows are not merely surplus — they make  $JM^{-1}J^T$  singular, and the constrained Newton solve has no answer to give. `posim` does no redundant-row elimination, so it reports the failure rather than guessing.

The fix is to brace with the *smallest thing that does the job*. A **CONSTRAIN** is one row:

$$5 + 4 + 1 = 10 \text{ rows on 12 freedoms.}$$

Two freedoms left, full rank, and the mechanism runs. Better still, the rod fixes the bend in closed form. The output shaft's centre stays 0.3 from the cross and 0.4243 from the post, so it rides the circle where those two spheres meet, sweeping a cone of half-angle  $\arctan(0.3/0.6) = 26.565^\circ$  about a line itself  $26.565^\circ$  off the  $x$  axis. The bend therefore runs from  $0$  to  $53.130^\circ$  and no further:

$$\cos 53.130^\circ = 0.6 \quad \text{exactly.}$$

Scrub the recording to the sharpest frame and the shafts are at 0.6000004. That number is never given to the integrator; it comes out because the hinge, the universal joint and the rod all held at once.

**The rule.** Before adding a joint, add up the rows it will bring and compare with the freedoms actually left. If the total meets or exceeds the freedoms, you have either locked the mechanism or made it singular, and a fatal return flag is the honest outcome either way.